
Causal-EPIG: Causally Aligned Active CATE Estimation

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Abstract

Estimating the Conditional Average Treatment Effect (CATE) is constrained by the high cost of obtaining outcome measurements, making active learning valuable. However, conventional strategies suffer from a fundamental objective mismatch: they reduce uncertainty in model parameters or observable outcomes rather than the unobservable causal quantities of interest. We address this via the principle of causal objective alignment, positing that acquisition functions should target potential outcomes or CATE directly. We operationalize this through Causal-EPIG, a framework adapting Expected Predictive Information Gain to quantify uncertainty reduction in causal quantities. We derive two distinct strategies: a comprehensive approach that targets the joint potential-outcome structure, and a focused approach that directly targets the CATE estimand for sample efficiency. We provide theoretical justification for our framework, establishing a formal link between CATE estimation error and posterior uncertainty in causal quantities. Extensive experiments demonstrate that our strategies improve sample efficiency over standard baselines, and crucially, reveal that the preferred strategy is context-dependent, contingent on the base estimator and treatment-effect structure. Our framework thus provides a principled guide for sample-efficient CATE estimation in practice.

1. Introduction

Understanding the causal effects of interventions is central to reliable decision-making in complex domains. Causal inference provides a principled framework for this purpose

by modeling the underlying dependencies in real-world data (Pearl, 2009; Hernán & Robins, 2010; Wager, 2024). Its importance is evident across domains such as health-care (Foster et al., 2011), economics (Heckman, 2000), and recommendation systems (Gao et al., 2024a), where accurately assessing the impact of actions is critical for designing effective policies and delivering personalized interventions. Estimating the Conditional Average Treatment Effect (CATE) is a key problem in this context, as it captures how treatment effects vary across individuals (Künzel et al., 2019). While randomized controlled trials remain the gold standard for causal inference, they are often impractical due to prohibitive costs and ethical barriers (Benson & Hartz, 2017). Consequently, researchers increasingly rely on observational data, which scale more readily but introduce the additional challenge of controlling for confounding to ensure valid causal conclusions (Imbens & Rubin, 2015).

Beyond the challenge of controlling for confounding, a critical practical constraint in observational studies is the acquisition of ground-truth outcome data. This typically requires a costly process, such as expert annotation or long-term patient follow-up, to obtain a reliable outcome for each subject (Nwankwo et al., 2025). Consequently, in many real-world scenarios, this process is expensive, logistically demanding, and subject to privacy or ethical restrictions (Gao et al., 2024b; Kallus & Mao, 2025). In healthcare, for example, measuring outcomes may require costly diagnostic tests or invasive procedures such as biopsies and large-scale tumor imaging, where the resulting label scarcity can severely impact the accuracy of CATE estimation (Bi et al., 2019; Wen et al., 2025). In economics and the social sciences, outcomes such as long-term income trajectories or behavioral changes often require extensive, costly follow-up (McKenzie, 2012). These resource constraints are further compounded when the study population differs systematically from the target population of interest (Kern et al., 2024). For instance, a health maintenance organization in California might need to rely on evidence from a study conducted years prior in Switzerland, whose participants fail to reflect the heterogeneity of the local population (Kallus et al., 2018). This challenge of generalizing findings across populations, formally known as ensuring external validity (Rothwell, 2005) or, more specifically, transportability (Bareinboim & Pearl, 2013), critically undermines the real-world utility of

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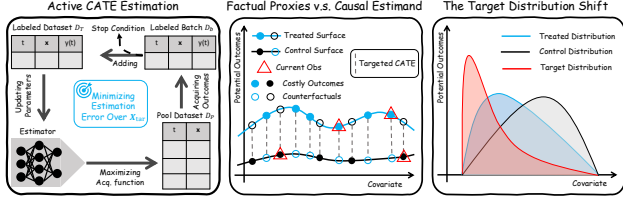


Figure 1. (Left) Illustrates the pool-based AL pipeline for CATE estimation. (Middle) Highlights the proxy-target disconnect: the goal is to learn the CATE, the data consist only of single factual outcomes. (Right) Shows the challenge of target distribution shift, where the sampling pool differs from the target population.

causal estimates. To address the dual challenges of resource scarcity and population shift, effective methods should be sample-efficient and target-aware, so that CATE estimates remain accurate for the population of interest.

Challenges. Active learning (AL) offers a principled framework for maximizing estimation accuracy under a fixed budget, yet its application to CATE estimation is hindered by a fundamental challenge. Standard AL methods are built for a world of factual observations, designed to reduce uncertainty about observable outcomes or model parameters. The objective of CATE estimation, however, is to precisely quantify an unobservable counterfactual difference. This misalignment between a fact-based acquisition process and a counterfactual-based goal can lead to inefficient data selection, as acquired outcomes may fail to reduce uncertainty where it matters most: in the treatment effect itself.

Existing literature has made valuable progress in adapting conventional AL paradigms for CATE estimation. Seminal works (Jesson et al., 2021) have explored criteria like factual outcome uncertainty or information gain about model parameters. A detailed discussion of related work is provided in App. A. While these criteria are natural and often effective, they remain indirect for CATE estimation: they optimize uncertainty in surrogate quantities rather than in the target causal estimand itself. This issue becomes more pronounced under distribution shift, when the sampling pool may not represent the target population. Consequently, a critical gap persists: *the need for causally aligned acquisition strategies that directly target treatment-effect uncertainty over the population of interest.*

Contributions. This paper makes the following contributions. *A New Principle.* We introduce the principle of Causal Objective Alignment (COA), arguing that the structural disconnect between observable data and the causal estimand mandates acquisition functions that are explicitly designed for the final causal goal (Sec. 3). *A Target-Aligned Information-Theoretic Framework.* We develop Causal-EPIG, an information-theoretic framework that operationalizes our principle (Sec. 4.1). From this unified framework, we derive two distinct, principled acquisition strategies: one that targets the joint potential-outcome struc-

ture, and a second that directly targets the final CATE estimand. *Broad Model Compatibility.* We demonstrate that Causal-EPIG is a flexible framework that naturally accommodates a range of popular Bayesian CATE estimators (Sec. 4.2), including Gaussian Process (GP)-based models like Causal Multi-task GP (Alaa & Van Der Schaar, 2017) and Non-Stationary GP (Alaa & Schaar, 2018), as well as the tree-based Bayesian Causal Forests (Hahn et al., 2020). *Theoretical Justification.* We provide a formal theoretical justification for our framework (Sec. 4.4), establishing a rigorous link between CATE estimation error and posterior uncertainty in causal quantities (Prop. 4.1), clarifying the relationship between prediction-oriented and parameter-based information targets (Prop. E.8), and providing a convergence analysis that bounds posterior uncertainty under a global greedy acquisition strategy (Thm. 4.3). *Extensive Empirical Validation.* We conduct comprehensive experiments showing that both strategies derived from our framework improve sample efficiency over a wide array of baselines (Sec. 5). Our main experiments and targeted ablations further show that the choice between the comprehensive and focused strategies reflects a context-dependent trade-off, governed by both the base estimator and how the treatment-effect signal is organized relative to the full outcome structure.

2. Preliminaries and Problem Setup

Potential Outcomes and CATE Estimation. Our analytical framework is grounded in the Neyman-Rubin Potential Outcomes (POs) model (Rubin, 2005). We define the random variables x , t , and y to represent the covariates, treatment, and outcome, respectively, with domains $\mathcal{X}$, $\{0, 1\}$, and $\mathcal{Y}$. We denote realizations by x , t , and y . The two potential outcomes are $y(0)$ and $y(1)$, corresponding to the outcome under control and treatment. The propensity score is $\pi(x) = p(t = 1 | x = x)$. Our primary goal is to estimate the CATE, defined as $\tau(x) := \mathbb{E}[y(1) - y(0) | x = x]$. To ensure identifiability, we impose the following assumptions.

Assumption 2.1. Unconfoundedness: Given the covariates x , treatment assignment t is independent of the potential outcomes, i.e., $(y(1), y(0)) \perp\!\!\!\perp t | x$. This reflects the assumption that there is no unmeasured confounding conditional on x .

Positivity (Common Support): For any covariates x in the support of the covariate distribution, the probability of receiving either treatment is non-zero: $0 < \pi(x) < 1$. **SUTVA (Stable Unit Treatment Value):** An individual’s potential outcomes are unaffected by the treatment assignments of others (No Interference), and the observed outcome is the potential outcome corresponding to the treatment received, i.e., $y = ty(1) + (1 - t)y(0)$ (Consistency).

Under Ass. 2.1, the CATE becomes identifiable as the difference in the conditional expectations of the observed outcome, which we denote $f(x, t) := \mathbb{E}[y | x = x, t = t]$. This is

expressed as:

$$\tau(\mathbf{x}) = \mathbb{E}[y|\mathbf{x} = \mathbf{x}, t = 1] - \mathbb{E}[y|\mathbf{x} = \mathbf{x}, t = 0]. \quad (1)$$

2.1. Pool-based active estimation of CATE

We begin with a cohort of factual covariate–treatment pairs. A random warm-start subset of size n_T is labeled to form $D_T = \{(\mathbf{x}_i, t_i, y_i)\}_{i=1}^{n_T}$; after removing these units, let $D_P = \{(\mathbf{x}_i, t_i)\}_{i=1}^{n_P}$ denote the remaining accessible unlabeled pool. The AL loop then proceeds iteratively: a model trained on the current D_T informs an acquisition function, which selects a batch of n_b instances from D_P to be labeled. These labeled instances are added to D_T , and the process repeats until the post-warm-start acquisition budget n_B is exhausted (Qin et al., 2021). Our objective is to learn a CATE model, $\hat{\tau}(\mathbf{x})$, that is accurate over a specific target distribution of interest, $p_{\text{tar}}(\mathbf{x})$, which may differ from the distribution of the sampling pool $p_{\text{pool}}(\mathbf{x})$. In the standard no-shift setting, this target distribution can coincide with the pool distribution; in transductive settings, it is represented by an unlabeled target covariate set. To formalize this, we evaluate performance using the square root of the Precision in Estimating Heterogeneous Effects ($\sqrt{\epsilon_{\text{PEHE}}}$) (Hill, 2011). This metric is defined as the root mean squared error over the target distribution and is empirically estimated using a finite target set $\mathbf{X}_{\text{tar}}$ drawn from $p_{\text{tar}}(\mathbf{x})$:

$$\sqrt{\epsilon_{\text{PEHE}}[\hat{\tau}]} := \sqrt{\mathbb{E}_{p_{\text{tar}}(\mathbf{x})} [(\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x}))^2]}. \quad (2)$$

Remark 2.2. Observational Constraint vs. Experimental Design. A key constraint in our setup is that we operate on observational data, even during acquisition. For any instance $(\mathbf{x}_i, t_i)$, we can only query its pre-existing factual outcome $y_i = y_i(t_i)$ and cannot intervene to assign a new treatment and observe the counterfactual. This limitation distinguishes our problem from adaptive experimental design, which requires the freedom to assign treatments (Toth et al., 2022; Kato et al., 2024; Cha & Lee, 2025; Klein et al., 2025; Zhang et al., 2025). This constraint is common in sensitive domains like healthcare and social sciences, where treatment assignment is governed by external factors. A more detailed discussion of the related literature on adaptive experimental design is provided in App. A.3.

✦ Key Objective. In this problem setup, the central challenge is to design a principled utility function, $U(\cdot)$, that quantifies the informativeness of any single candidate data point $(\mathbf{x}, t)$ from the pool. The acquisition strategy is then to select the candidate, denoted $(\mathbf{x}_s, t_s)$, that is deemed most valuable by maximizing this function:

$$(\mathbf{x}_s, t_s) = \arg \max_{(\mathbf{x}, t) \in D_P} U(\mathbf{x}, t \mid D_T, \mathbf{X}_{\text{tar}}). \quad (3)$$

While this defines the selection of a single instance, this process is typically extended to the batch setting by greedily selecting the n_b candidates that yield the highest utility scores.

3. Aligning AL with Causal Objectives

This section analyzes the structure of active outcome acquisition for CATE estimation, highlighting an objective mismatch with standard AL paradigms. We show that this mismatch motivates a core principle, Causal Objective Alignment (COA), for designing principled and sample-efficient acquisition strategies.

From Indirect Proxies to the Causal Estimand. In standard AL, the path from query to knowledge is direct. The learning objective is aligned with the data-generating process: one queries a point $\mathbf{x}_i$ to observe a label y_i , which is a direct (though noisy) signal for the target function $f(\mathbf{x}_i)$. Direct applications of this paradigm to CATE estimation simply adopt these standard targets: they might focus on the uncertainty of the observable response surface, $f(\mathbf{x}, t)$, or on the uncertainty of the model’s internal parameters, θ . However, this can create an objective mismatch for CATE estimation, as illustrated in Fig. 1. Both the data we can acquire (factual outcomes) and the model’s parameters are indirect proxies for our true inferential goal. This goal is to estimate causal quantities that are not directly observed: the two potential-outcome response surfaces, $f(\mathbf{x}, 0)$ and $f(\mathbf{x}, 1)$, and the CATE function, $\tau(\mathbf{x})$, derived from them as in Eq. 1. This disconnect between proxy acquisition targets and the causal quantities of interest motivates our core design principle:

Principle 3.1 (Causal Objective Alignment). *An effective acquisition strategy for active CATE estimation should be causally aligned. Its utility function should quantify the value of a query with respect to causal quantities of interest, such as potential-outcome response surfaces or the CATE itself, rather than only indirect proxies such as factual outcomes or model parameters.*

The COA principle also makes the target population explicit: utility is evaluated relative to $\mathbf{X}_{\text{tar}}$, the covariates on which CATE accuracy is required. This naturally frames active CATE estimation as a transductive learning problem (a connection detailed in App. A.2.1). It also reveals a spectrum of acquisition strategies, ranging from approaches that target indirect proxies (e.g., factual uncertainty or parameter uncertainty) to causally aligned strategies that target the joint potential-outcome surfaces or the CATE itself. The importance of this alignment is amplified under distribution shift ($p_{\text{tar}}(\mathbf{x}) \neq p_{\text{pool}}(\mathbf{x})$), where the sampling pool may not represent the target population. This perspective provides the conceptual foundation for the Causal-EPIG framework

introduced next.

4. Active CATE Estimation via Causal-EPIG

We operationalize the COA principle through the Causal-EPIG framework. We derive two strategies representing a trade-off between comprehensiveness and focus, arguing that the preferred choice depends on the estimator and how the treatment-effect signal is organized. Following their formal definitions, we demonstrate the framework’s compatibility with Bayesian CATE models. Here $p_{\text{pool}}(\mathbf{x}, t)$ denotes the empirical joint distribution of factual covariate–treatment pairs in the accessible observational pool, so t^* is the factual treatment attached to a sampled pool unit.

4.1. The Causal-EPIG Acquisition Function

The design of Causal-EPIG is best motivated by a direct contrast with these standard AL criteria, as illustrated in Tab. 1. As defined above, methods like EIG/BALD are *parameter-focused*, aiming to reduce uncertainty over the model parameters (θ). This objective is indirect; reducing global parameter uncertainty does not guarantee a targeted reduction in CATE uncertainty (Jesson et al., 2021). This limitation can persist even for causal adaptations. For instance, in models like BCF that adopt a separable structure, $f(\mathbf{x}, t) = \mu(\mathbf{x}) + t \cdot \tau(\mathbf{x})$, with parameters $\theta = (\theta_\mu, \theta_\tau)$, one could target the CATE parameters θ_τ specifically. This is a more causally aligned parameter-space objective, but it still focuses on the model’s internal representation rather than the target causal predictions themselves (Fawkes et al., 2025). In restricted finite-dimensional settings where $\tau(\mathbf{x})$ is a direct linear readout of θ_τ , parameter- and prediction-focused objectives can be closely related; beyond this regime, non-linear dependence and target-population weighting can induce different acquisition rankings. Standard EPIG (Smith et al., 2023) elevates the objective by targeting future predictions, but in its usual form these predictions remain factual outcomes. This is not fully aligned with CATE, which is inherently a comparative quantity, $\tau(\mathbf{x}) = \mathbb{E}[y(1) - y(0) \mid \mathbf{x}]$. A data point that is highly informative for one potential outcome may offer little information about the other, and thus may not efficiently reduce uncertainty about their difference.

✚ **A Comprehensive Strategy: Targeting the Joint PO Structure (Causal-EPIG- μ).** A direct application of our COA principle is to target the two latent conditional potential-outcome responses at a target covariate. Let $\mathbf{f}_{\text{PO}}(\mathbf{x}^*) := (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1))$, where $f(\mathbf{x}^*, t) = \mathbb{E}[y(t) \mid \mathbf{x} = \mathbf{x}^*]$ under Ass. 2.1. Targeting the joint posterior of $\mathbf{f}_{\text{PO}}(\mathbf{x}^*)$ retains the modeled dependence between the two response surfaces. This leads to our Potential Outcome-based (PO-based) strategy:

$$\text{Causal-EPIG-}\mu(\mathbf{x}, t) := \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{I}(y; \mathbf{f}_{\text{PO}}(\mathbf{x}^*) \mid \mathbf{x}^*, D'_T)]. \quad (4)$$

Here the targets are latent conditional response values, while y denotes the future factual noisy observation obtained by querying $(\mathbf{x}, t)$. By seeking data that reduce uncertainty over this joint distribution, Causal-EPIG- μ encourages learning the arm-specific outcome surfaces from which the CATE is derived. This can be advantageous when treatment-effect information is distributed across the two surfaces, but it may also spend acquisition budget on prognostic variation that is less relevant to their contrast. For completeness, we also discuss a simpler, additive variant in App. D.2, which approximates this broader objective.

✚ **A Focused Strategy: Directly Targeting the Causal Estimand (Causal-EPIG- τ).** In contrast, an alternative approach is to focus the acquisition objective directly on the final inferential goal: the CATE function $\tau(\mathbf{x}^*)$. This focused strategy can be more sample-efficient when the treatment-effect signal is sufficiently learnable and concentrated in the differenced quantity, by prioritizing data points that directly resolve uncertainty in this causal estimand. Formally, we define the Causal-EPIG- τ utility as the expected IG about the CATE:

$$\text{Causal-EPIG-}\tau(\mathbf{x}, t) := \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{I}(y; \tau(\mathbf{x}^*) \mid \mathbf{x}^*, D'_T)]. \quad (5)$$

The mutual information represents the expected reduction in CATE posterior entropy. An equivalent and computationally useful formulation uses the KL divergence to frame this utility as the expected belief update about the CATE, $\tau(\mathbf{x}^*)$, after a potential observation y . Intuitively, a high utility score signifies that an observation at $(\mathbf{x}, t)$ is expected to significantly change our beliefs about the CATE in the target population, marking it as a highly informative candidate.

Positioning and Trade-off. Causal-EPIG differs from Causal-BALD-style criteria (Jesson et al., 2021) in its information target. Whereas methods such as τ -BALD measure how much a target prediction reveals about model parameters, Causal-EPIG measures how much a future factual observation reveals about target causal predictions. Thus, informativeness is evaluated in prediction space rather than parameter space. Our practical utilities use a mean-marginal aggregation over $\mathbf{X}_{\text{tar}}$, estimating the target expectation by a finite sum; a global alternative would compute mutual information with the full vector of target predictions, e.g., $\mathbb{I}(y; \boldsymbol{\tau})$, thereby capturing inter-target dependencies at higher computational cost (Hübner et al., 2024). Within Causal-EPIG, the μ - and τ -based utilities reflect a comprehensiveness–focus trade-off: Causal-EPIG- μ preserves the joint potential-outcome structure, while Causal-EPIG- τ targets the differenced estimand directly. The preferred choice is therefore context-dependent, varying with the estimator and with how treatment-effect information is organized relative to the full outcome structure.

Table 1. Comparison of information-theoretic acquisition functions for active CATE estimation. For brevity, D'_T denotes the context after conditioning on the candidate covariates and treatment, $D'_T = D_T \cup \{(x, t)\}$, before observing the random outcome y .

	Non-Causal-Aware	Causal-Aware
EIG	$I(y; \theta \mid D'_T)$ (μ -BALD)	$I(y; \theta_\tau \mid D'_T)$ (Causal-EIG)
EPIG	$\mathbb{E}_{p_{\text{pool}}(x^*, t^*)} [I(y; y^*(t^*) \mid (x^*, t^*), D'_T)]$	$\mathbb{E}_{p_{\text{tar}}(x^*)} [I(y; (f(x^*, 0), f(x^*, 1)) \mid x^*, D'_T)]$ (PO-based) $\mathbb{E}_{p_{\text{tar}}(x^*)} [I(y; \tau(x^*) \mid x^*, D'_T)]$ (CATE-based)

4.2. Realization with Bayesian CATE Estimators

Causal-EPIG is estimator-agnostic at the level of the acquisition target, but its implementation requires access to predictive uncertainty over the relevant causal quantities. We outline realization strategies for two major classes of Bayesian CATE estimators.

Exact Realization with GP Models. For CATE estimators based on GPs, such as CMGP (Alaa & Van Der Schaar, 2017) and NSGP (Alaa & Schaar, 2018), the joint posterior predictive distribution over any set of points is, by construction, a multivariate Gaussian. Consequently, the required predictive variances and covariances can be extracted directly from the GP’s analytical posterior covariance matrix. In this setting, the mutual information has an exact closed-form solution. For example, in the scalar jointly Gaussian case,

$$I(a; b) = \frac{1}{2} \log \frac{\text{Var}[a] \text{Var}[b]}{\text{Var}[a] \text{Var}[b] - \text{Cov}[a, b]^2}. \quad (6)$$

This allows for an efficient and exact implementation of Causal-EPIG with GP-based models.

Approximate Realization for General Bayesian Models.

For more complex models whose posterior predictive distribution is represented by samples, such as the MCMC output of Bayesian regression trees (Hill, 2011) and BCF (Hahn et al., 2020), exact mutual information is generally unavailable. To make Causal-EPIG tractable for this broad class of models, we employ a computationally efficient Gaussian approximation, following prior work (Kirsch, 2023; Jesson et al., 2021). This strategy involves forming paired posterior predictive samples of the candidate outcome and the target causal quantity, and then fitting a multivariate Gaussian to these paired draws. The mean vector and covariance matrix are estimated empirically from the set of n_M posterior samples. For instance, when applying this to BCF, the covariance term is computed from its MCMC draws as

$$\text{Cov}[y, \tau(x^*)] \approx \text{Cov}(\{f(x, t \mid \theta_j)\}_{j=1}^{n_M}, \{\tau(x^* \mid \theta_j)\}_{j=1}^{n_M}). \quad (7)$$

Once this approximation is made, we can reuse the closed-form solution for mutual information (Eq. 6). This approach provides a practical recipe for pairing Causal-EPIG with sample-based estimators while avoiding expensive nested MC simulations.

4.3. Budgeted Acquisition Algorithm

The Causal-EPIG utility serves as our acquisition function for ranking candidate data points. We employ this utility in an iterative active learning strategy designed to estimate the CATE under a fixed budget (Qin et al., 2021; Jesson et al., 2021). This process is illustrated conceptually in Fig. 1 and detailed in Alg. 1 (App. C.1). It begins with a warm-start phase where a small random batch of data is labeled. Subsequently, in each round, the algorithm computes the Causal-EPIG utility for all candidates in the unlabeled pool. A batch of points with the highest utility scores is selected, their outcomes are queried, and the CATE model is retrained on the expanded labeled set. This cycle continues until the budget is exhausted. The batch size controls the usual adaptivity–computation trade-off: smaller batches allow more frequent model updates, while larger batches reduce retraining cost.

4.4. Theoretical Analysis

We provide a theoretical justification for the Causal-EPIG framework. We first show that, under a Bayesian model-based squared-error criterion, active CATE estimation reduces to minimizing posterior CATE variance. We then analyze posterior-uncertainty convergence for the global PO-based Causal-EPIG objective, and discuss the scope of this guarantee for the practical mean-marginal variant.

4.4.1. OBJECTIVE: CATE ERROR AND POSTERIOR CATE VARIANCE

The following result connects CATE estimation error to a tractable model-based criterion and shows that the relevant uncertainty is the posterior variance of the CATE, which depends on the joint posterior structure of the two potential outcomes. Here, ϵ_{PEHE}^M denotes the squared PEHE model risk, not its square root.

Proposition 4.1. Assume $f(x, 0)$ and $f(x, 1)$ are modeled by a joint GP with fixed hyperparameters and Gaussian observation likelihood, and the CATE estimator is the posterior mean $\hat{\tau}_s(x) = \mathbb{E}_s[\tau(x)]$. For one-step active CATE estimation under this model-based criterion, minimizing the expected model-based estimation error, $\mathbb{E}_{s+1}[\epsilon_{\text{PEHE}}^M(\hat{\tau}_{s+1})]$, is equivalent under the GP assumption to minimizing the

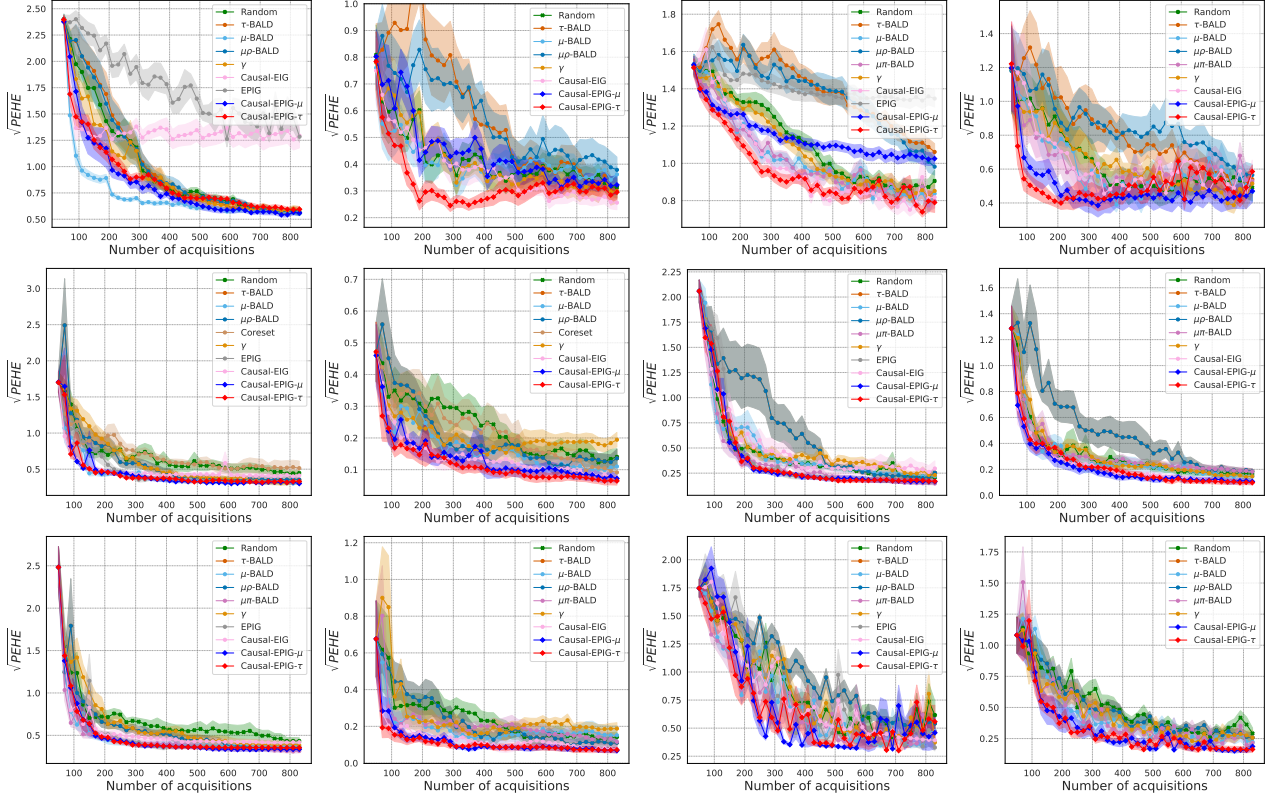


Figure 2. Comparison of $\sqrt{\text{PEHE}}$ for BCF, CMGP, and NSGP (rows). The first two columns show results on CausalBALD, while the last two correspond to Hahn, covering both regular and distributional shift settings.

integrated posterior CATE variance:

$$\arg \min_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{s+1}[\epsilon_{\text{PEHE}}^M(\hat{\tau}_{s+1})] \equiv \arg \min_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{\text{Ptar}(\mathbf{x}^*)}[\text{Var}_{s+1}[\tau(\mathbf{x}^*)]]. \quad (8)$$

Here, $\text{Var}_{s+1}[\tau(\mathbf{x}^*)]$ is the posterior variance of the Bayesian random variable $\tau(\mathbf{x}^*)$, which explicitly retains the joint posterior structure:

$$\begin{aligned} \text{Var}_{s+1}[\tau(\mathbf{x}^*)] &= \text{Var}_{s+1}[f(\mathbf{x}^*, 1)] + \text{Var}_{s+1}[f(\mathbf{x}^*, 0)] \\ &\quad - 2 \text{Cov}_{s+1}(f(\mathbf{x}^*, 1), f(\mathbf{x}^*, 0)). \end{aligned} \quad (9)$$

A detailed proof is provided in App. E.8. This result clarifies why Causal-EPIG should target predictive uncertainty in causal quantities rather than uncertainty in parameters alone. In particular, the CATE variance depends not only on the marginal uncertainties of the two outcome surfaces, but also on their posterior covariance. This motivates acquisition functions that retain the joint potential-outcome structure.

4.4.2. CONVERGENCE OF POSTERIOR UNCERTAINTY

Prop. 4.1 shows that reducing posterior CATE variance is the natural model-based objective. For convergence analysis, we focus on the global PO-based variant, Causal-EPIG- μ -G, which jointly targets all potential outcomes over $\mathbf{X}_{\text{tar}}$. We map this objective to transductive active learning

(TAL) (Hübötter et al., 2024) by viewing the two potential-outcome surfaces as a single GP $f(\tilde{\mathbf{x}})$ on the augmented space $\tilde{\mathcal{X}} = \mathcal{X} \times \{0, 1\}$. This induces an augmented target space $\tilde{\mathbf{X}}_{\text{tar}}$, containing target potential-outcome components, and an augmented pool $\tilde{\mathcal{D}}_P$ of factual covariate–treatment pairs. The Causal-EPIG- μ -G utility then coincides with the Global ITL objective on the augmented instance, while our acquisition rule retains the pool restriction that each factual unit is queried at most once (App. E.1).

Let D_0 denote a warm-start set of size n_T , let D_n be the labeled set after n post-warm-start active acquisitions, and write $n_P := |\tilde{\mathcal{D}}_P|$ for the size of the accessible pool at the start of this acquisition phase. All mutual information and variance terms below are computed under the posterior GP after conditioning on D_0 . For $1 \leq k \leq n_P$, define the finite-pool information capacity

$$\begin{aligned} \gamma_k &:= \max_{\substack{\tilde{A} \subseteq \tilde{\mathcal{D}}_P \\ |\tilde{A}| \leq k}} \text{I}(f_{\tilde{\mathbf{X}}_{\text{tar}}}; y_{\tilde{A}} | D_0), \\ \eta_{\tilde{\mathcal{D}}_P}^2(\tilde{\mathbf{x}}^*) &:= \text{Var}[f(\tilde{\mathbf{x}}^*) | D_0, y_{\tilde{\mathcal{D}}_P}], \end{aligned} \quad (10)$$

where $y_{\tilde{\mathcal{D}}_P}$ denotes the noisy outcomes that would be obtained by querying every accessible pool unit once. Thus, $\eta_{\tilde{\mathcal{D}}_P}^2$ is the posterior uncertainty attainable after exhausting the finite pool. The analysis uses the following submodular-ity condition.

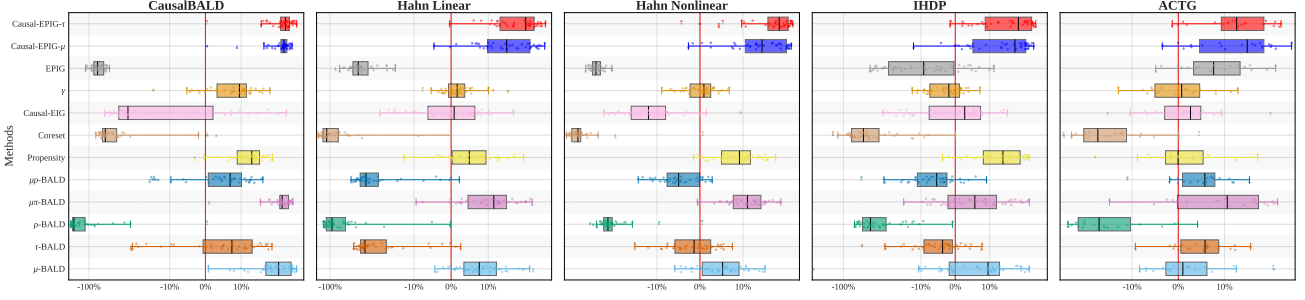


Figure 3. Average relative improvement of acquisition functions over *Random acquisition* on five datasets.

Assumption 4.2. The Causal-EPIG- μ -G set utility $\psi_{\tilde{\mathcal{X}}_{\text{tar}}}(\tilde{A}) := \mathbb{I}(f_{\tilde{\mathcal{X}}_{\text{tar}}}; y_{\tilde{A}} | D_0)$, $\tilde{A} \subseteq \tilde{\mathcal{D}}_P$, is submodular.

This assumption is used only for the global PO-based convergence analysis. It should not be read as a universal submodularity claim for all Causal-EPIG variants; in fully transductive settings, or for derived-target and mean-marginal objectives, synergistic effects may occur. We discuss its scope in App. E.2.

For the result below, the GP kernel, its hyperparameters, and the Gaussian observation-noise model are fixed during acquisition, with strictly positive noise variances. Data acquisition follows sequential, no-replacement Causal-EPIG- μ -G with deterministic tie-breaking, and n_B denotes the post-warm-start acquisition budget.

Theorem 4.3. *Under Ass. 4.2, there exists a finite, instance-dependent constant $C > 0$, independent of n_B , such that, for every $1 \leq n_B \leq n_P$ and $\tilde{\mathbf{x}}^* \in \tilde{\mathcal{X}}_{\text{tar}}$,*

$$\text{Var}[f(\tilde{\mathbf{x}}^*) | D_{n_B}] \leq \eta_{\mathcal{D}_P}^2(\tilde{\mathbf{x}}^*) + C \frac{\gamma_{n_B}}{\sqrt{n_B}}. \quad (11)$$

This compact finite-budget statement is proved without repeatable access. App. E.3 gives an explicit admissible C and a sharper budget-dependent contraction bound that reaches $\eta_{\mathcal{D}_P}^2$ exactly when the pool is exhausted. Because C may depend on the fixed finite pool and its information-capacity profile, the display should not be read as a repeatable-sampling asymptotic rate. The appendix also gives a per-batch analogue for stale top- n_b selection using the *global* Causal-EPIG- μ -G utility; it does not cover top- n_b selection under the mean-marginal utilities used in our experiments.

Mean-marginal objective. The global objective is theoretically convenient but computationally expensive. Our experiments therefore use the mean-marginal utilities in Eqs. 4–5, which are not covered by the global theorem or its global-utility batch analogue. For the sequential Causal-EPIG- τ rule, App. E.5 shows under a monotonicity condition that the maximal mean-marginal information gain is at most $\bar{\gamma}_n^\tau/n$. It also gives the exact Gaussian identity linking the information collected about each scalar CATE target to its posterior-variance contraction. These statements

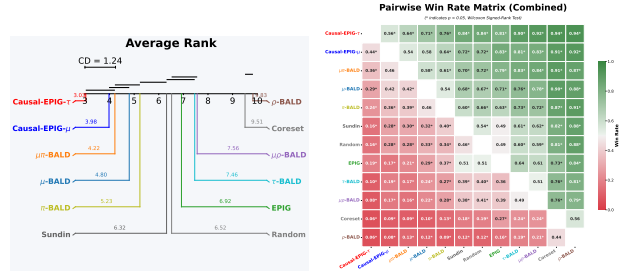


Figure 4. Average comparison of acquisition functions across all synthetic datasets: (a) critical difference diagram, and (b) pairwise win-rate matrix with statistical significance.

do not imply a uniform average-variance convergence rate for the practical mean-marginal objective; obtaining such a rate, and extending the result to stale-score batches, requires additional analysis.

5. Experimental Results

To assess the sample efficiency of our Causal-EPIG framework, we conduct extensive experiments on several benchmarks. These include synthetic datasets based on the data-generating processes from Causal-BALD (Jesson et al., 2021) and Hahn et al. (2020), as well as two well-established semi-synthetic benchmarks: the Infant Health and Development Program (IHDP) (Hill, 2011) and AIDS Clinical Trials Group Study 175 (ACTG-175) (Hammer et al., 1996). Full details regarding the dataset characteristics and partitioning for each benchmark are available in App. B.

Base Bayesian CATE Estimators, Baselines, and Metrics.

To demonstrate the flexibility of our framework, we implement Causal-EPIG with three distinct and well-established Bayesian CATE estimators: BCF, CMGP, and NSGP. These models are natural partners for our information-theoretic acquisition functions, as they provide the necessary posterior uncertainty over the relevant causal quantities. For brevity, the main text focuses on these primary models; comprehensive results for all setups, including an additional estimator from the Causal-BALD study, are provided in App. F. For our baselines, we compare against a range of acquisition functions, including Random, γ -acquisition

(S-type error rate control) (Sundin et al., 2019), coresets selection (Qin et al., 2021), Causal-EIG (Fawkes et al., 2025), and the suite of methods from Causal-BALD (Jes-son et al., 2021). Detailed implementations for all methods are available in App. C. Our primary evaluation metric is the Root PEHE ($\sqrt{\hat{\text{PEHE}}}$ or $\sqrt{\text{PEHE}}$ for short; Eq. 2), computed on the target set $\mathbf{X}_{\text{tar}}$. All results are reported as the mean and standard deviation across 10 independent runs. In addition to performance curves, we report the relative Root PEHE improvement over the Random baseline for a holistic summary of sample efficiency. This metric is calculated at each acquisition step k as $(\sqrt{\text{PEHE}}_{\text{Random}}(k) - \sqrt{\text{PEHE}}_{\text{Method}}(k)) / \sqrt{\text{PEHE}}_{\text{Random}}(k)$. Finally, we aggregate these point-wise improvements across all steps to visualize the distribution of performance gains for each method, offering insight into its consistency throughout the active learning process.

5.1. Synthetic Data

Results. Fig. 2 shows that Causal-EPIG improves sample efficiency on synthetic benchmarks. The focused strategy, **Causal-EPIG- τ** (red), achieves leading or competitive sample efficiency across base estimators (BCF, CMGP, NSGP) and distribution-shift settings. The comprehensive strategy, **Causal-EPIG- μ** (blue), also significantly outperforms most baselines, though its effectiveness is slightly attenuated with BCF. One possible explanation is BCF’s separation of prognostic (μ) and treatment (τ) effects, which may make joint potential-outcome targeting less direct. Fig. 3 shows that Causal-EPIG- τ yields the highest average improvement, supporting the efficiency of direct targeting when the CATE signal is well captured by the estimator. Statistical analysis of 180 dataset-seed combinations (Fig. 4) corroborates these findings. In the Critical Difference (CD) diagram, Causal-EPIG- τ achieves the best average rank (3.03), followed by Causal-EPIG- μ (3.98); both rank ahead of the remaining baselines under the Nemenyi comparison (Nemenyi test, $\text{CD}=1.24$, $\alpha = 0.05$). Pairwise Wilcoxon signed-rank tests further confirm that Causal-EPIG- τ significantly outperforms 11 of 11 baselines ($p < 0.05$) with an 80.6% win rate, while Causal-EPIG- μ beats 8 baselines with a 72.1% win rate. Notably, neither variant significantly loses to any baseline. Comprehensive results and ablation studies on stability (varying initializations, pool sizes, batch sizes, and the Deep-GP estimator) are provided in App. F.1, F.2.

Mechanistic comparison of Causal-EPIG- μ and Causal-EPIG- τ . To clarify the trade-off between the two Causal-EPIG targets, we run a targeted synthetic study varying the treatment-effect formation mechanism and the predictive share of the active informative block, $\rho = \frac{k_{\text{pred}}}{k_{\text{prog}} + k_{\text{pred}}}$, while keeping the total active informative dimension fixed, following the predictive/prognostic composition used in re-

lated active feature-acquisition studies (Piskorz et al., 2025). Causal-EPIG- μ preserves the full joint potential-outcome structure, whereas Causal-EPIG- τ compresses the target to the differenced quantity. Thus, the relevant factor is not merely the number of predictive variables, but how the treatment-effect signal is organized relative to the full outcome structure. Fig. 5 shows two regimes. In the *surface-split* setting, the effect is induced by differences between arm-specific outcome surfaces, so learning the full PO pair remains beneficial and Causal-EPIG- μ is favored. In the *cancellation* settings, part of the full-outcome structure cancels or becomes weakly relevant in the treatment-effect difference, making direct CATE targeting more efficient; the relative advantage therefore shifts toward Causal-EPIG- τ as ρ increases.

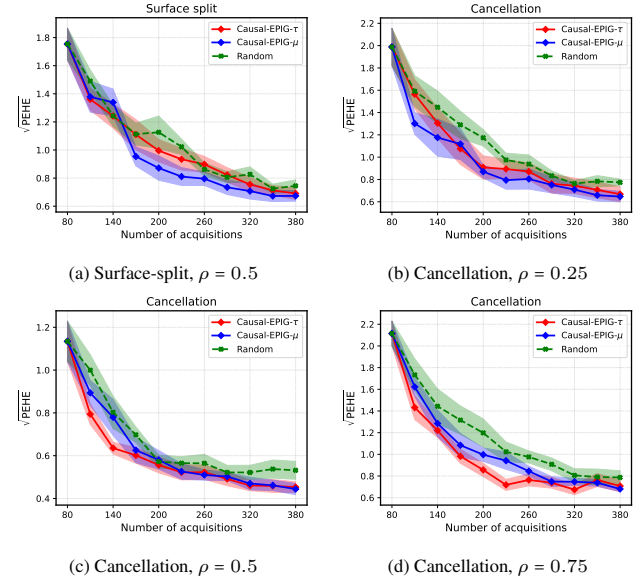


Figure 5. Targeted synthetic comparison of Causal-EPIG- μ and Causal-EPIG- τ . Surface-split favors Causal-EPIG- μ , while cancellation regimes increasingly favor Causal-EPIG- τ as the predictive share ρ grows.

Computational Considerations. The sample efficiency of Causal-EPIG comes at the cost of a more computationally intensive acquisition function than simpler baselines. This is a standard computational-statistical trade-off. All main experiments use the mean-marginal formulation, whose acquisition cost scales linearly with the pool size, target-set size, and number of predictive samples. The approach is therefore most suitable when outcome acquisition is expensive relative to model scoring, such as in clinical or industrial studies. In such settings, the additional computation can be justified by improved sample efficiency. A detailed breakdown of per-sample runtimes is provided in App. D.3.

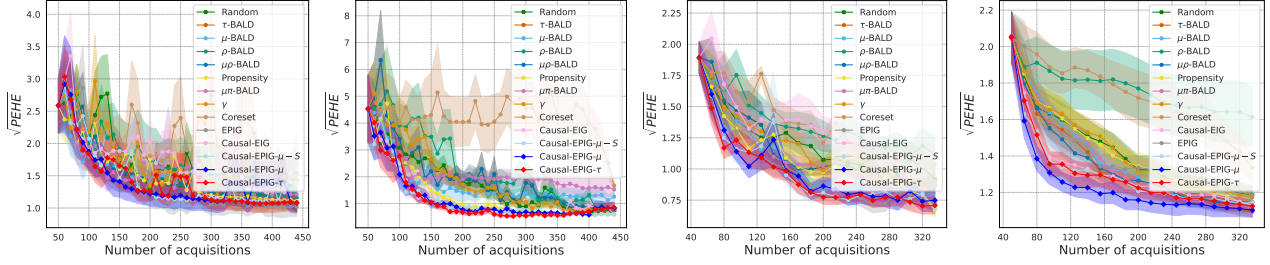


Figure 6. Comparison of $\sqrt{\text{PEHE}}$ on IHDP and ACTG-175 datasets. We evaluate the following four experimental settings: (1) CMGP on IHDP, (2) CMGP on IHDP under target shift, (3) CMGP on ACTG-175, and (4) BCF on ACTG-175.

5.2. Semi-synthetic Data

Benchmarks. We evaluate our framework on two well-established semi-synthetic benchmarks designed to mimic challenges of real-world observational studies. The first, the IHDP (Hill, 2011), simulates selection bias by removing a non-random subset of the treated group from a randomized trial. The second, ACTG-175 (Hammer et al., 1996), constructs an observational cohort by excluding participants based on their enrollment symptoms.

Results and Analysis. The results on these semi-synthetic benchmarks (Fig. 6 and Fig. 3) further support the practical effectiveness of Causal-EPIG and illustrate the trade-off between its two strategies. Across both the IHDP and ACTG datasets, both Causal-EPIG- μ and Causal-EPIG- τ achieve competitive or leading performance, demonstrating the overall strength of our causally-aligned, prediction-focused approach. The strong performance of the comprehensive Causal-EPIG- μ strategy is particularly noteworthy, suggesting that its more comprehensive targeting of the joint potential-outcome structure is highly effective in these complex, lower signal-to-noise settings. Furthermore, these results underscore our thesis that the optimal acquisition strategy is context-dependent. On the IHDP benchmark, which is defined by significant selection bias, we observe that propensity-based baselines also perform competitively. This is consistent with IHDP’s strong selection-bias structure, where propensity-aware criteria are naturally competitive. Overall, the key advantage of Causal-EPIG is not that one variant is universally dominant, but that the framework provides two target-aligned acquisition rules whose relative strengths vary with the estimator and data-generating structure. Full results are provided in App. F.1.

6. Discussions

This work addressed the objective mismatch between standard active learning and CATE estimation by introducing the principle of causal objective alignment. We operationalized this principle with the Causal-EPIG framework, a flexible information-theoretic approach that yields two distinct acquisition strategies: a comprehensive strategy targeting

the joint potential-outcome structure and a focused strategy targeting the CATE itself. Our experiments show that both target-aligned strategies generally improve sample efficiency over strong baselines, and further support our central hypothesis that the choice between them reflects a context-dependent trade-off. This finding provides empirical evidence for our principle: aligning the acquisition objective with the causal target is important, but the preferred form of alignment depends on the estimator and data-generating structure. By providing a framework that navigates this trade-off, our work supports more cost-effective and reliable CATE estimation where outcome acquisition is expensive.

Limitations and Future Work. Causal-EPIG assumes no unobserved confounding and relies on the base estimator’s predictive uncertainty, especially its joint uncertainty over the relevant causal quantities. This uncertainty may be unreliable in very low-data regimes or under severe model misspecification (Zhang et al., 2025). While estimator-agnostic at the acquisition-target level, its effectiveness is bounded by the underlying CATE estimator and by the quality of its uncertainty estimates. Future work includes integrating robust models such as CausalPFN (Balazadeh et al., 2025), addressing hidden confounding (Li et al., 2023), and extending the framework to adaptive experimental design.

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Impact Statement

This work aims to make CATE estimation more sample-efficient when outcome acquisition is costly, with potential benefits in resource-constrained domains such as personalized medicine and development economics. By accounting for target populations, Causal-EPIG may also help improve external validity under population shift. Its deployment, however, still depends on valid causal assumptions: reducing CATE uncertainty does not ensure correctness under model misspecification or unobserved confounding.

References

- Addanki, R., Arbour, D., Mai, T., Musco, C., and Rao, A. Sample constrained treatment effect estimation. In *Advances in Neural Information Processing Systems*, volume 35, pp. 5417–5430, 2022.
- Alaa, A. and Schaar, M. Limits of estimating heterogeneous treatment effects: Guidelines for practical algorithm design. In *International Conference on Machine Learning*, pp. 129–138. PMLR, 2018.
- Alaa, A. M. and Van Der Schaar, M. Bayesian inference of individualized treatment effects using multi-task gaussian processes. In *Advances in neural information processing systems*, volume 30, 2017.
- Balazadeh, V., Kamkari, H., Thomas, V., Li, B., Ma, J., Cresswell, J. C., and Krishnan, R. CausalPFN: Amortized causal effect estimation via in-context learning. In *ICML 2025 Workshop on Scaling Up Intervention Models*, 2025.
- Bareinboim, E. and Pearl, J. A general algorithm for deciding transportability of experimental results. *Journal of causal Inference*, 1(1):107–134, 2013.
- Benson, K. and Hartz, A. J. A comparison of observational studies and randomized, controlled trials. In *Research ethics*, pp. 213–221. Routledge, 2017.
- Bi, W. L., Hosny, A., Schabath, M. B., Giger, M. L., Birkbak, N. J., Mehrtash, A., Allison, T., Arnaout, O., Abbosh, C., Dunn, I. F., et al. Artificial intelligence in cancer imaging: clinical challenges and applications. *CA: a cancer journal for clinicians*, 69(2):127–157, 2019.
- Cha, T. and Lee, D. Abc3: Active bayesian causal inference with cohn criteria in randomized experiments. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 26769–26777, 2025.
- Cook, T., Mishler, A., and Ramdas, A. Semiparametric efficient inference in adaptive experiments. In *Causal Learning and Reasoning*, pp. 1033–1064. PMLR, 2024.
- Efron, B. Forcing a sequential experiment to be balanced. *Biometrika*, 58(3):403–417, 1971.
- Fawkes, J., Ter-Minassian, L., Ivanova, D. R., Shalit, U., and Holmes, C. C. Is merging worth it? securely evaluating the information gain for causal dataset acquisition. In *International Conference on Artificial Intelligence and Statistics*, pp. 1423–1431. PMLR, 2025.
- Foster, J. C., Taylor, J. M., and Ruberg, S. J. Subgroup identification from randomized clinical trial data. *Statistics in medicine*, 30(24):2867–2880, 2011.
- Gao, C., Zheng, Y., Wang, W., Feng, F., He, X., and Li, Y. Causal inference in recommender systems: A survey and future directions. *ACM Transactions on Information Systems*, 42(4):1–32, 2024a.
- Gao, E. and Sejdinovic, D. ActiveCQ: Active estimation of causal quantities. In *The Fourteenth International Conference on Learning Representations*, 2026.
- Gao, E., Bondell, H., Huang, W., and Gong, M. A variational framework for estimating continuous treatment effects with measurement error. In *The Twelfth International Conference on Learning Representations*, 2024b.
- Gao, E., Zhang, L., Fawkes, J., Zuo, A., Liu, W., Li, H., Gong, M., and Sejdinovic, D. Observationally informed adaptive causal experimental design. *arXiv preprint arXiv:2603.03785*, 2026.
- Ghadiri, M., Arbour, D., Mai, T., Musco, C., and Rao, A. B. Finite population regression adjustment and non-asymptotic guarantees for treatment effect estimation. *Advances in Neural Information Processing Systems*, 36: 74180–74212, 2023.
- Hahn, J., Hirano, K., and Karlan, D. Adaptive experimental design using the propensity score. *Journal of Business & Economic Statistics*, 29(1):96–108, 2011.
- Hahn, P. R., Murray, J. S., and Carvalho, C. M. Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15(3):965–1056, 2020.
- Hammer, S. M., Katzenstein, D. A., Hughes, M. D., Gundacker, H., Schooley, R. T., Haubrich, R. H., Henry, W. K., Lederman, M. M., Phair, J. P., Niu, M., et al. A trial comparing nucleoside monotherapy with combination therapy in hiv-infected adults with cd4 cell counts from 200 to 500 per cubic millimeter. *New England Journal of Medicine*, 335(15):1081–1090, 1996.
- Harshaw, C., Sävje, F., Spielman, D. A., and Zhang, P. Balancing covariates in randomized experiments with the gram–schmidt walk design. *Journal of the American Statistical Association*, 119(548):2934–2946, 2024.
- Heckman, J. J. Causal parameters and policy analysis in economics: A twentieth century retrospective. *The Quarterly Journal of Economics*, 115(1):45–97, 2000.
- Hernán, M. A. and Robins, J. M. *Causal inference*. CRC Boca Raton, FL, 2010.
- Hill, J. L. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011.

- Hübottter, J., Treven, L., As, Y., and Krause, A. Transductive active learning: Theory and applications. In *Advances in Neural Information Processing Systems*, volume 37, pp. 124686–124755, 2024.
- Imbens, G. W. and Rubin, D. B. *Causal inference in statistics, social, and biomedical sciences*. Cambridge university press, 2015.
- Jesson, A., Tigas, P., van Amersfoort, J., Kirsch, A., Shalit, U., and Gal, Y. Causal-bald: Deep bayesian active learning of outcomes to infer treatment-effects from observational data. In *Advances in Neural Information Processing Systems*, volume 34, pp. 30465–30478, 2021.
- Kallus, N. and Mao, X. On the role of surrogates in the efficient estimation of treatment effects with limited outcome data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):480–509, 2025.
- Kallus, N., Puli, A. M., and Shalit, U. Removing hidden confounding by experimental grounding. *Advances in neural information processing systems*, 31, 2018.
- Kallus, N., Mao, X., and Zhou, A. Interval estimation of individual-level causal effects under unobserved confounding. In *The 22nd international conference on artificial intelligence and statistics*, pp. 2281–2290. PMLR, 2019.
- Kato, M., Ishihara, T., Honda, J., and Narita, Y. Efficient adaptive experimental design for average treatment effect estimation. *arXiv preprint arXiv:2002.05308*, 2020.
- Kato, M., McAlinn, K., and Yasui, S. The adaptive doubly robust estimator and a paradox concerning logging policy. *Advances in neural information processing systems*, 34: 1351–1364, 2021.
- Kato, M., Oga, A., Komatsubara, W., and Inokuchi, R. Active adaptive experimental design for treatment effect estimation with covariate choice. In *International Conference on Machine Learning*. PMLR, 2024.
- Kern, C., Kim, M. P., and Zhou, A. Multi-accurate cate is robust to unknown covariate shifts. *Transactions on Machine Learning Research*, 2024.
- Kirsch, A. Black-box batch active learning for regression. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. Expert Certification.
- Klein, O. N., Hüyük, A., Shamir, R., Shalit, U., and van der Schaar, M. Towards regulatory-confirmed adaptive clinical trials: Machine learning opportunities and solutions. In *International Conference on Artificial Intelligence and Statistics*, pp. 4969–4977. PMLR, 2025.
- Krantsevich, N., He, J., and Hahn, P. R. Stochastic tree ensembles for estimating heterogeneous effects. In *International Conference on Artificial Intelligence and Statistics*, pp. 6120–6131. PMLR, 2023.
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the national academy of sciences*, 116(10):4156–4165, 2019.
- Li, F., Ding, P., and Mealli, F. Bayesian causal inference: a critical review. *Philosophical Transactions of the Royal Society A*, 381(2247):20220153, 2023.
- MacKay, D. J. Information-based objective functions for active data selection. *Neural Computation*, 4(4):590–604, 1992.
- McKenzie, D. Beyond baseline and follow-up: The case for more t in experiments. *Journal of development Economics*, 99(2):210–221, 2012.
- Nwankwo, E., Goldkind, L., and Zhou, A. Batch-adaptive annotations for causal inference with complex-embedded outcomes. *arXiv preprint arXiv:2502.10605*, 2025.
- Pearl, J. *Causality*. Cambridge university press, 2009.
- Piskorz, J., Astorga, N., Berrevoets, J., and van der Schaar, M. Active feature acquisition for personalised treatment assignment. In *International Conference on Artificial Intelligence and Statistics*, pp. 4330–4338. PMLR, 2025.
- Qin, T., Wang, T.-Z., and Zhou, Z.-H. Budgeted heterogeneous treatment effect estimation. In *International Conference on Machine Learning*, pp. 8693–8702. PMLR, 2021.
- Rothwell, P. M. External validity of randomised controlled trials: “to whom do the results of this trial apply?”. *The Lancet*, 365(9453):82–93, 2005.
- Rubin, D. B. Causal inference using potential outcomes: Design, modeling, decisions. *Journal of the American Statistical Association*, 100(469):322–331, 2005.
- Settles, B. Active learning literature survey. 2009.
- Smith, F. B., Kirsch, A., Farquhar, S., Gal, Y., Foster, A., and Rainforth, T. Prediction-oriented bayesian active learning. In *International Conference on Artificial Intelligence and Statistics*, pp. 7331–7348. PMLR, 2023.
- Sundin, I., Schulam, P., Siivola, E., Vehtari, A., Saria, S., and Kaski, S. Active learning for decision-making from imbalanced observational data. In *International conference on machine learning*, pp. 6046–6055. PMLR, 2019.

- Tabord-Meehan, M. Stratification trees for adaptive randomisation in randomised controlled trials. *Review of Economic Studies*, 90(5):2646–2673, 2023.
- Toth, C., Lorch, L., Knoll, C., Krause, A., Pernkopf, F., Peharz, R., and Von Kügelgen, J. Active bayesian causal inference. *Advances in Neural Information Processing Systems*, 35:16261–16275, 2022.
- van der Laan, M. J. The construction and analysis of adaptive group sequential designs. 2008.
- Wager, S. Causal inference: A statistical learning approach, 2024.
- Wen, H., Chen, T., Gong, M., Chai, L. K., Sadiq, S., and Yin, H. Enhancing treatment effect estimation via active learning: A counterfactual covering perspective. In *International Conference on Machine Learning*, 2025.
- Zhang, Z., Wang, H., Li, H., and Lin, Z. Active treatment effect estimation via limited samples. In *Forty-second International Conference on Machine Learning*, 2025.

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A. Additional Related Works

A.1. Active CATE Estimation

Our work addresses *active outcome acquisition for CATE estimation*: treatment assignments are observational, but outcome measurements are costly (Nwankwo et al., 2025). The goal is to select which outcomes to query from an existing cohort in order to improve a CATE model most efficiently. Existing approaches have mainly adapted standard active learning heuristics. One line uses diversity-based sampling, such as coresets selection, to cover the covariate space (Qin et al., 2021; Wen et al., 2025). Another targets specific causal error types rather than the overall CATE estimation error (Sundin et al., 2019). These methods are useful, but their acquisition objectives remain indirect, relying on geometric diversity or auxiliary error criteria rather than uncertainty in the target causal estimand. More closely related are information-theoretic causal acquisition rules. Causal-EIG (Fawkes et al., 2025) targets information gain about CATE-specific parameters (θ_τ), while Causal-BALD (Jesson et al., 2021) measures how much a causal prediction informs the model parameters. More broadly, Gao & Sejdinovic (2026) study active estimation of general causal quantities beyond CATE, deriving acquisition rules from posterior uncertainty of causal functionals. These works are causal-aware, but they either operate through parameter-space objectives or target broader causal functionals rather than the CATE-specific predictive quantities considered here. In contrast, Causal-EPIG builds on prediction-oriented Bayesian active learning (Smith et al., 2023) and defines informativeness directly with respect to target causal predictions, such as potential outcomes or the CATE itself. This yields acquisition rules aligned with the causal estimand over the target population, rather than only with geometric coverage, auxiliary error criteria, or parameter uncertainty.

A.2. Inductive and Transductive Goals in Active Learning

Active learning is often framed as an inductive problem: the goal is to train a model that generalizes to future unseen data under an IID assumption (Settles, 2009). Transductive active learning instead optimizes performance on a specified target set. Pool-based AL has elements of both: acquisition decisions are made using a finite unlabeled pool, but the usual goal is still to learn a broadly generalizable model. This distinction becomes important under distribution shift, where the sampling pool distribution p_{pool} may differ from the target distribution p_{tar} . Recent target-aware acquisition strategies formalize this transductive perspective (MacKay, 1992; Hübötter et al., 2024; Smith et al., 2023).

A.2.1. CONNECTION BETWEEN ACTIVE CATE ESTIMATION AND TAL

Active CATE estimation can be viewed as a transductive problem because acquisition should be evaluated with respect to a target population $\mathbf{X}_{\text{tar}}$. In TAL (Hübötter et al., 2024), the learner queries from a sample space $\mathcal{S}$ to reduce uncertainty on a target space $\mathcal{A}$. In our setting, $\mathbf{X}_{\text{tar}}$ plays the role of the target set: the goal is not uniform accuracy over the whole covariate space, but accurate CATE estimation on this population. The CATE setting also has an additional structural feature. A query at $(\mathbf{x}_i, t_i)$ reveals only one factual outcome, providing information about the corresponding response surface $f(\mathbf{x}_i, t_i)$. The inferential target, however, is the contrast $\tau(\mathbf{x}_i) = f(\mathbf{x}_i, 1) - f(\mathbf{x}_i, 0)$, which depends on a pair of potential-outcome response surfaces. Thus, even when $p_{\text{pool}}(\mathbf{x}) = p_{\text{tar}}(\mathbf{x})$, acquisition remains target-dependent: the utility of a factual observation should be judged by how much it reduces uncertainty in the unobserved causal quantities over $\mathbf{X}_{\text{tar}}$. Under distribution shift, this transductive aspect is further strengthened because the learner must use observations from p_{pool} to improve CATE estimates on p_{tar} . This perspective motivates the Causal Objective Alignment principle in Sec. 3 and the Causal-EPIG framework in Sec. 4.

A.3. Adaptive Experimental Design

A separate line of causal active learning studies active or adaptive experimental design, where the learner can influence treatment assignment. One major direction studies adaptive randomization, where treatment probabilities are updated to improve efficiency for estimators such as the ATE (van der Laan, 2008; Hahn et al., 2011). Recent work refines these designs using online nuisance estimates, martingale arguments, and specialized estimators for adaptively collected data (Kato et al., 2021; Tabord-Meehan, 2023; Kato et al., 2020; Cook et al., 2024). A second direction studies finite-pool design, including classical covariate-balance trade-offs (Efron, 1971) and modern sampling methods based on leverage scores or balancing walks (Addanki et al., 2022; Ghadiri et al., 2023; Harshaw et al., 2024). A third direction uses Bayesian experimental design for treatment-effect estimation, including methods with regulatory constraints (Klein et al., 2025), structural uncertainty from causal discovery (Toth et al., 2022), GP-based CATE variance minimization (Cha & Lee, 2025), finite-sample guarantees in network settings (Zhang et al., 2025), and designs that use observational data as a prior for prospective trials (Gao et al.,

2026). Our setting is different. Adaptive experimental design asks *who should receive which treatment*; active outcome acquisition asks *whose outcome should be measured*, given that treatment assignments are already fixed by observational or practical constraints. This distinction is important in domains such as healthcare, where treatments may already have been assigned but outcome measurements, such as biopsies or genetic sequencing, remain costly. Nevertheless, both settings allocate limited resources to reduce causal uncertainty, suggesting that target-aware acquisition principles may also inform future adaptive experimental design.

B. Datasets

We evaluate Causal-EPIG on two fully synthetic benchmarks and two semi-synthetic benchmarks based on IHDP and ACTG-175. In all settings, the outcome-generation process is known, so the ground-truth CATE is available for PEHE evaluation. We also include covariate-shift variants for the CausalBALD, Hahn, and IHDP datasets. **General protocol.** For synthetic benchmarks, we generate a pool set D_P of 2000 instances, a validation set of 200 instances, and a test set of 2000 instances. Unless otherwise stated, active learning starts from 50 randomly labeled instances. For CausalBALD and Hahn, we run 40 acquisition rounds with batch size 20; for IHDP, 40 rounds with batch size 10; and for ACTG-175, 20 rounds with batch size 15. Results are averaged over 10 independent trials. In the standard setting, the acquisition target set is the pool covariates, $\mathbf{X}_{\text{tar}} = \mathbf{X}_P$, and evaluation is reported on both pool and held-out test covariates. In the covariate-shift setting, the acquisition target set is the shifted test set, $\mathbf{X}_{\text{tar}} = \mathbf{X}_{\text{test}}$, and test performance is the primary metric.

B.1. CausalBALD Synthetic Dataset

We use a one-dimensional synthetic dataset adapted from CausalBALD (Jesson et al., 2021), which itself builds on Kallus et al. (2019). In the standard setting, $x \sim \mathcal{N}(0, 1)$, $t \mid x \sim \text{Bern}(\pi(x))$, and $\pi(x) = \text{sigmoid}(2x + 0.5)$. The potential-outcome means are $\mu_0(x) = 1 + 2\sin(2x)$ and $\mu_1(x) = 2x + 3 - 2\sin(2x)$, with additive noise $\epsilon \sim \mathcal{N}(0, 1)$. Hence, $\tau(x) = \mu_1(x) - \mu_0(x) = 2x + 2 - 4\sin(2x)$. For the covariate-shift setting, training and pool covariates remain drawn from $\mathcal{N}(0, 1)$, while test/target covariates are drawn from $\mathcal{U}(0.2, 0.5)$. The outcome functions are unchanged.

B.2. Hahn Synthetic Dataset

We use the simulation design from Hahn et al. (2020), with covariates $x \in \mathbb{R}^5$. The first three covariates are Gaussian, $x_1, x_2, x_3 \sim \mathcal{N}(0, 1)$; $x_4 \sim \text{Bernoulli}(0.5)$; and $x_5 \sim \mathcal{U}\{1, 2, 3\}$. We use the nonlinear prognostic function and heterogeneous treatment-effect function: $\mu(x) = -6 + g(x_5) + 6|x_3 - 1|$ and $\tau(x) = 1 + 2x_2x_4$, where $g(1) = 2$, $g(2) = -1$, and $g(3) = -4$. We use a modified propensity score to create a more challenging observational setting. Let $\tilde{\mu}(x) = 3\mu(x)/\sigma_\mu$, where σ_μ is the population standard deviation of $\mu(x)$. We define $\pi(x) = 0.8\phi(\tilde{\mu}(x)) - 0.5x_1 + \xi$, with $\xi \sim \mathcal{U}(0.05, 0.15)$, and draw $t \sim \text{Bernoulli}(\pi(x))$. The observed outcome is $y = \mu(x) + t\tau(x) + \epsilon$, with Gaussian noise scaled to signal-to-noise ratio 3. For the covariate-shift setting, the pool is generated as above, while the test continuous covariates are drawn from $x_1, x_2, x_3 \sim \mathcal{U}(0.2, 0.5)$. The discrete covariates and outcome functions remain unchanged.

B.3. IHDP Semi-Synthetic Dataset

We use the Infant Health and Development Program (IHDP) semi-synthetic benchmark (Hill, 2011). The dataset contains 747 subjects, with 139 treated and 608 controls, and 25 covariates: 6 continuous and 19 binary. We split the data into 523 pool/training subjects and 224 test subjects, and standardize continuous covariates. In the standard setting, a sparse coefficient vector β_B is generated by sampling each entry from $\{0.0, 0.1, 0.2, 0.3, 0.4\}$ with probabilities $\{0.6, 0.1, 0.1, 0.1, 0.1\}$. The potential-outcome means are $\mu_0(x) = \exp((x + 0.5)\beta_B)$ and $\mu_1(x) = (x + 0.5)\beta_B - \omega_B$, where ω_B is chosen so that the ATT equals 4. Potential outcomes are generated as $y_0(x) = \mu_0(x) + \epsilon$ and $y_1(x) = \mu_1(x) + \epsilon$, with $\epsilon \sim \mathcal{N}(0, 1)$. For the covariate-shift setting, the first two continuous test covariates, birth weight and head circumference, are resampled from $\mathcal{U}(0, 0.5)$. The coefficient vector is sampled as above, but the first two coefficients are set to zero. The outcome functions are $\mu_0(x) = \exp((x + 0.5)\beta_B)$ and $\mu_1(x) = \exp((x + 0.5)\beta_B) + 3x_{\text{bw}}x_{\text{b.head}}$, so that $\tau(x) = 3x_{\text{bw}}x_{\text{b.head}}$.

B.4. ACTG-175 Semi-Synthetic Dataset

We use the AIDS Clinical Trials Group Study 175 (ACTG-175) dataset (Hammer et al., 1996). Following the semi-synthetic setup, an observational cohort is formed from the randomized trial by removing a non-random subset of patients, specifically those not showing symptomatic HIV infection. The resulting dataset contains 813 subjects, with 281 treated and 532

Algorithm 1 Budgeted Batch Active Learning for CATE Estimation

Require: Factual candidate cohort D_{all} , target set X_{tar} , utility function U , warm-start size n_T , batch size n_b , post-warm-start acquisition budget n_B .

Ensure: Final labeled set D_T and final CATE estimator $\hat{\tau}(\cdot)$.

- 1: Initialize labeled set $D_T \leftarrow \emptyset$ and candidate pool $D_P \leftarrow D_{\text{all}}$.
// – Warm-start Phase –
 - 2: Select an initial random batch $D_{\text{init}} \subset D_P$ of size n_T .
 - 3: Query factual outcomes and form labeled batch $\tilde{D}_{\text{init}} = \{(\mathbf{x}, t, y) : (\mathbf{x}, t) \in D_{\text{init}}\}$.
 - 4: Update $D_T \leftarrow D_T \cup \tilde{D}_{\text{init}}$ and $D_P \leftarrow D_P \setminus D_{\text{init}}$.
 - 5: Train initial CATE estimator $\hat{\tau}(\cdot)$ on D_T .
 - 6: Set the post-warm-start acquisition count $n \leftarrow 0$.
// – Main Active Learning Loop –
 - 7: **while** $n < n_B$ **and** $D_P \neq \emptyset$ **do**
 - 8: Compute $U(\mathbf{x}_i, t_i \mid D_T, X_{\text{tar}})$ for each $(\mathbf{x}_i, t_i) \in D_P$.
 - 9: Set $m_b \leftarrow \min\{n_b, n_B - n, |D_P|\}$ and select a batch D_b of m_b candidates according to the acquisition rule.
 - 10: Query factual outcomes for all $(\mathbf{x}, t) \in D_b$.
 - 11: Form $\tilde{D}_b = \{(\mathbf{x}, t, y) : (\mathbf{x}, t) \in D_b\}$.
 - 12: Update $D_T \leftarrow D_T \cup \tilde{D}_b$, $D_P \leftarrow D_P \setminus D_b$, and $n \leftarrow n + m_b$.
 - 13: Retrain or update $\hat{\tau}(\cdot)$ on D_T .
 - 14: **end while**
 - 15: **return** $D_T, \hat{\tau}(\cdot)$.
-

controls, and 12 covariates: 3 continuous and 9 binary. We split the data into 569 pool/training subjects and 244 test subjects, standardize continuous covariates, and simulate outcomes using nonlinear functions with interactions. The prognostic score and CATE are

$$\begin{aligned} \mu(x) &= 6 + 0.3x_{\text{wtkg}}^2 - \sin(x_{\text{age}})(x_{\text{gender}} + 1) + 0.6x_{\text{hemo}}x_{\text{race}} - 0.2x_{\text{z30}}, \\ \tau(x) &= 1 + 1.5 \sin(x_{\text{wtkg}})(x_{\text{karnof_hi}} + 1) + 2x_{\text{age}}. \end{aligned} \quad (12)$$

We set $\mu_0(x) = \mu(x)$ and $\mu_1(x) = \mu(x) + \tau(x)$. Potential outcomes are generated as $y_0(x) = \mu_0(x) + \epsilon$ and $y_1(x) = \mu_1(x) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$ and $\sigma_y = \frac{\max(\mu) - \min(\mu)}{8}$. The observed outcome is $y = (1 - t)y_0(x) + ty_1(x)$.

C. Model and Acquisition Details

This section summarizes the active CATE acquisition loop, the acquisition functions used in our experiments, and the base CATE estimators. Additional hyperparameter details are provided in the released code.

C.1. Active CATE Estimation Loop

Alg. 1 describes the batch active-learning protocol used throughout our experiments. Starting from a random warm-start set, each round scores candidates in the remaining pool, acquires a batch of factual outcomes, augments the labeled set, and retrains or updates the CATE estimator.

Batch acquisition. For all acquisition functions, candidates are first scored individually. For baselines adapted from Causal-BALD (Jesson et al., 2021), we follow the original stochastic softmax acquisition protocol. For Causal-EPIG, we use deterministic selection of up to the top n_b candidates, which was empirically comparable to temperature-based stochastic variants while being simpler and faster.

C.2. Acquisition Functions

We compare Causal-EPIG with several active acquisition baselines. Unless otherwise stated, each full batch consists of the n_b candidates with the highest utility scores.

Random. Random acquisition selects up to n_b candidates uniformly at random from the remaining pool D_P .

Causal-BALD variants. We include the acquisition functions from Causal-BALD (Jesson et al., 2021), which adapt BALD

to causal quantities through parameter-space information gains. The main variants are:

- **τ -BALD**: mutual information between the CATE-related quantity and model parameters, e.g., $I(y(1) - y(0); \theta \mid D'_T)$.
- **μ -BALD**: mutual information between the relevant potential outcome and model parameters, e.g., $I(y(t); \theta \mid D_T \cup \{(x, t)\})$.
- **Propensity**: an acquisition rule targeting uncertainty in the propensity score $\pi(x)$.
- **Combined variants**: weighted combinations such as $\mu\pi$ -BALD and $\mu\rho$ -BALD.

The key distinction is that Causal-BALD is parameter-focused, whereas Causal-EPIG evaluates informativeness in the space of target causal predictions. We use the official implementation from the authors.¹

Sign-ambiguity BALD. Following Sundin et al. (Sundin et al., 2019) and the adaptation in Causal-BALD (Jesson et al., 2021), this baseline targets points where the sign of the CATE is most uncertain. For posterior draw k , let $\tau_k(x) = \mu_{1,k}(x) - \mu_{0,k}(x)$ and let $s_k^2(x)$ be the sum of the arm-wise conditional predictive variances. Define

$$q_k(x) := \Phi\left(-\frac{|\tau_k(x)|}{s_k(x)}\right), \quad (13)$$

where Φ is the standard normal CDF. In our standardized implementation, both arm-wise conditional variances are one, so $s_k(x) = \sqrt{2}$. The utility is approximated as

$$U_{\text{sign}}(x) := H(\text{Bernoulli}(\bar{q}(x))) - \frac{1}{K} \sum_{k=1}^K H(\text{Bernoulli}(q_k(x))), \quad (14)$$

where $\bar{q}(x) = K^{-1} \sum_{k=1}^K q_k(x)$.

Coreset selection (QHTE). We implement the coreset acquisition strategy from QHTE (Qin et al., 2021).² The pool is partitioned into treatment arms, D_P^1 and D_P^0 , and coreset scores are computed separately within each arm. For a candidate (x, t) , the utility is

$$U_{\text{QHTE}}(x, t) := \min_{x' \in X_T^t} d(x, x'), \quad t \in \{0, 1\}, \quad (15)$$

where $X_T^t := \{x_i : (x_i, t_i, y_i) \in D_T, t_i = t\}$ is the set of labeled covariates in arm t , and $d(\cdot, \cdot)$ is derived from posterior predictive covariance when available. Scores from the two arms are then combined to select the top batch.

Causal-EIG. Causal-EIG (Fawkes et al., 2025) was proposed for prospective causal effect estimation, where it scores the information gain that a prospective dataset provides about causal parameters. We adapt it to pool-based AL by treating each candidate (x, t) as a size-one prospective dataset:

$$U_{\text{Causal-EIG}}(x, t) := I(y; \theta_\tau \mid x, t, D_T). \quad (16)$$

We implement this acquisition function with BCF and CMGP using the official code.³

EPIG. EPIG (Smith et al., 2023) measures how informative a candidate label is about future labels. In our setting, the factual EPIG baseline is

$$U_{\text{EPIG}}(x, t) := \mathbb{E}_{p_{\text{pool}}(x^*, t^*)} [I(y; y^*(t^*) \mid (x^*, t^*), D'_T)]. \quad (17)$$

Here $p_{\text{pool}}(x, t)$ denotes the empirical joint distribution of factual covariate–treatment pairs in the accessible observational pool; in particular, t^* is the factual treatment attached to the sampled pool unit. Unlike Causal-EPIG, this target remains a factual prediction rather than a causal estimand.

Causal-EPIG. Our proposed Causal-EPIG utilities replace the factual prediction target in EPIG with causal quantities. Causal-EPIG- μ targets the joint latent conditional-response pair $(f(x^*, 0), f(x^*, 1))$, while Causal-EPIG- τ targets the CATE $\tau(x^*) = f(x^*, 1) - f(x^*, 0)$. The main definitions are given in Sec. 4.

¹<https://github.com/OATML/causal-bald>

²<https://github.com/Qcer17/QHTE>

³https://github.com/LucileTerminassian/causal_prospective_merge

C.3. Base CATE Estimators

Bayesian Causal Forests. BCF models the conditional outcome as $\mathbb{E}[y \mid \mathbf{x}, t] = \mu(\mathbf{x}) + t\tau(\mathbf{x})$ using Bayesian tree ensembles (Hahn et al., 2020). We use its accelerated extension, XBCF (Krantsevich et al., 2023), with the reparameterization

$$f_\theta(\mathbf{x}, t) = a \tilde{\mu}_{\text{bcf}}(\mathbf{x}) + b_t \tilde{\tau}_{\text{bcf}}(\mathbf{x}), \quad (18)$$

so that the CATE is

$$\tau(\mathbf{x}) = (b_1 - b_0) \tilde{\tau}_{\text{bcf}}(\mathbf{x}). \quad (19)$$

Posterior CATE samples are obtained from MCMC draws. For information-based acquisition, we require joint predictive samples of candidate outcomes and target causal quantities. For each posterior draw s , we compute paired samples such as $(f^{(s)}(\mathbf{x}, t), \tau^{(s)}(\mathbf{x}^*))$ or $(f^{(s)}(\mathbf{x}, t), y^{*(s)}(0), y^{*(s)}(1))$, and approximate their joint distribution with a multivariate Gaussian estimated from the empirical mean and covariance of the samples (Kirsch, 2023; Jesson et al., 2021).

Causal Multitask Gaussian Processes. CMGP models the two potential outcome functions jointly as a multitask GP (Alaa & Van Der Schaar, 2017). It places a GP prior on $[f_0(\mathbf{x}), f_1(\mathbf{x})]^\top$ using a matrix-valued kernel, typically constructed through a Linear Model of Coregionalization. Given Gaussian likelihoods, the posterior over the potential outcomes is Gaussian. The CATE posterior is therefore also Gaussian:

$$\tau(\mathbf{x}_*) \sim \mathcal{N}(e^\top \boldsymbol{\mu}_{\text{post}}(\mathbf{x}_*), e^\top \boldsymbol{\Sigma}_{\text{post}}(\mathbf{x}_*, \mathbf{x}_*) e), \quad e = [-1, 1]^\top. \quad (20)$$

Thus, the mutual information quantities required by Causal-EPIG can be computed directly from the GP posterior covariance.

Non-Stationary Gaussian Processes. NSGP defines a GP over the augmented input space $\mathcal{X} \times \{0, 1\}$, treating treatment as an input (Alaa & Schaar, 2018). The model uses a non-stationary kernel allowing the control and treatment surfaces to have different smoothness properties. Posterior inference for $\tau(\mathbf{x}_*) = f(\mathbf{x}_*, 1) - f(\mathbf{x}_*, 0)$ follows from the joint Gaussian posterior over the two potential outcome values. As with CMGP, Causal-EPIG utilities are computed from the analytical posterior covariance.

Deep Kernel Learning estimator. For ablation, we also include the DUE estimator used in Causal-BALD (Jesson et al., 2021). DUE combines neural feature learning with a sparse variational GP. We use it to test whether Causal-EPIG remains effective beyond the main BCF/GP estimators.

C.4. Gaussian Approximation Diagnostics for BCF

For BCF, Causal-EPIG uses a Gaussian approximation to the joint posterior samples needed for mutual information computation. To assess this approximation, we test the joint posterior of $(\hat{y}, \hat{\tau})$ across five DGPs: CausalBALD, Hahn linear, Hahn nonlinear, IHDP, and ACTG. As expected, formal tests reject exact multivariate normality in all settings, but the magnitude of deviation varies. The semi-synthetic datasets show more moderate deviations, with Mardia kurtosis values closer to the bivariate Gaussian value of 8, while the synthetic datasets show heavier tails. We therefore view the Gaussian approximation as a tractable covariance-based approximation rather than an exact posterior assumption.

Table 2. Multivariate normality diagnostics for the joint posterior of $(\hat{y}, \hat{\tau})$ under BCF.

Dataset	Q-Q corr	χ^2 GoF p-value	HZ stat	Mardia kurtosis
CausalBALD	0.974	$< 10^{-10}$	0.210	11.430
Hahn (linear)	0.981	$< 10^{-10}$	0.189	9.655
Hahn (nonlinear)	0.993	$< 10^{-10}$	0.194	9.878
IHDP	0.989	1.45×10^{-3}	0.173	8.868
ACTG	0.989	6.94×10^{-3}	0.167	8.407

D. Interpretations and Derivations

This section provides derivations and additional interpretation for the information-theoretic acquisition functions used in the paper. We focus on Causal-EPIG- τ , since it is the most direct estimand-targeted objective. The derivation for Causal-EPIG- μ is identical except that the target variable is the joint latent conditional-response vector $(f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1))$ rather than the scalar CATE.

D.1. Derivation and Estimation of Causal-EPIG

Information-gain form. For a candidate query $(\mathbf{x}, t)$ and target covariate $\mathbf{x}^*$, let y be the unobserved factual outcome and let $\tau(\mathbf{x}^*)$ be the target CATE. The information gain from observing y is the posterior entropy reduction

$$\text{IG}_\tau((\mathbf{x}, t), y, \mathbf{x}^*) = H(\tau(\mathbf{x}^*) \mid D_T) - H(\tau(\mathbf{x}^*) \mid D_T \cup \{(\mathbf{x}, t, y)\}). \quad (21)$$

Averaging over the unknown outcome and target covariates gives

$$\text{Causal-EPIG}_\tau(\mathbf{x}, t) := \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} \mathbb{E}_{p(y|\mathbf{x}, t, D_T)} [\text{IG}_\tau((\mathbf{x}, t), y, \mathbf{x}^*)] \quad (22)$$

$$= \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} \mathbb{E}_{p(y, \tau(\mathbf{x}^*)|\mathbf{x}, t, D_T)} \left[\log \frac{p(\tau(\mathbf{x}^*) \mid D_T, y, \mathbf{x}, t)}{p(\tau(\mathbf{x}^*) \mid D_T)} \right] \quad (23)$$

$$= \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} \mathbb{E}_{p(y, \tau(\mathbf{x}^*)|\mathbf{x}, t, D_T)} \left[\log \frac{p(y, \tau(\mathbf{x}^*) \mid \mathbf{x}, t, D_T)}{p(y \mid \mathbf{x}, t, D_T) p(\tau(\mathbf{x}^*) \mid D_T)} \right] \quad (24)$$

$$= \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\text{I}(y; \tau(\mathbf{x}^*) \mid (\mathbf{x}, t), D_T)]. \quad (25)$$

This is the definition used in Eq. 5. Equivalently, Causal-EPIG can be interpreted as the expected KL divergence between the posterior and prior predictive distributions of the target causal quantity after observing the candidate outcome.

Model-specific realization. For GP-based CATE estimators such as CMGP and NSGP, the required joint predictive distributions are Gaussian, so the mutual information terms can be computed analytically from posterior variances and covariances. For sample-based estimators such as BCF, we form paired posterior samples of the candidate outcome and target causal quantity, fit a Gaussian using empirical moments, and then use the same closed-form MI expression. This avoids nested Monte Carlo while retaining the covariance structure needed by the acquisition utility.

D.1.1. GAUSSIAN CLOSED FORM

Assume that, conditional on D_T , the candidate outcome and target CATE have a bivariate Gaussian posterior

$$(y, \tau(\mathbf{x}^*) \mid \mathbf{x}, t, \mathbf{x}^*, D_T) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad (26)$$

with

$$\boldsymbol{\Sigma} = \begin{pmatrix} \text{Var}[y] & \text{Cov}[y, \tau(\mathbf{x}^*)] \\ \text{Cov}[\tau(\mathbf{x}^*), y] & \text{Var}[\tau(\mathbf{x}^*)] \end{pmatrix}. \quad (27)$$

Using the Gaussian entropy formula, $H(\mathbf{z}) = \frac{1}{2} \log((2\pi e)^d |\boldsymbol{\Sigma}_{\mathbf{z}}|)$, the mutual information is

$$\text{I}(y; \tau(\mathbf{x}^*)) = H(y) + H(\tau(\mathbf{x}^*)) - H(y, \tau(\mathbf{x}^*)) \quad (28)$$

$$= \frac{1}{2} \log \left(\frac{\text{Var}[y] \text{Var}[\tau(\mathbf{x}^*)]}{\det(\boldsymbol{\Sigma})} \right) \quad (29)$$

$$= \frac{1}{2} \log \left(\frac{\text{Var}[y] \text{Var}[\tau(\mathbf{x}^*)]}{\text{Var}[y] \text{Var}[\tau(\mathbf{x}^*)] - \text{Cov}[y, \tau(\mathbf{x}^*)]^2} \right). \quad (30)$$

Thus, for a finite target set $\mathbf{X}_{\text{tar}}$,

$$\text{Causal-EPIG}_\tau(\mathbf{x}, t) \approx \frac{1}{|\mathbf{X}_{\text{tar}}|} \sum_{\mathbf{x}^* \in \mathbf{X}_{\text{tar}}} \frac{1}{2} \log \left(\frac{\text{Var}[y] \text{Var}[\tau(\mathbf{x}^*)]}{\text{Var}[y] \text{Var}[\tau(\mathbf{x}^*)] - \text{Cov}[y, \tau(\mathbf{x}^*)]^2} \right), \quad (31)$$

where the variances and covariance are computed separately for each candidate–target pair.

D.2. A Taxonomy of Information-Theoretic Acquisition Functions

Causal-EPIG is part of a broader family of information-theoretic acquisition functions. Tab. 3 summarizes the main distinctions: parameter- vs. prediction-targeted objectives, factual vs. causal prediction targets, and mean-marginal vs. global formulations.

The first distinction is the information target. EIG-style methods, including Causal-EIG and Causal-BALD, are parameter-focused: they reduce uncertainty in θ_τ or θ . EPIG/ITL-style methods are prediction-focused: they reduce uncertainty

Table 3. A taxonomy of information-theoretic acquisition functions for active CATE estimation. The table distinguishes methods by their target of information gain and by whether they use a mean-marginal or global formulation.

Family	Target of Information Gain	Mean-Marginal Formulation	Global / Full Formulation
EIG	Model Parameters (θ)	$I(y; \theta_\tau (\mathbf{x}, t), D_T)$ or $I(y; \theta (\mathbf{x}, t), D_T)$	
EPIG / ITL	Factual Outcome (y^*)	$\mathbb{E}_{p_{\text{pool}}(\mathbf{x}^*, t^*)}[I(y; y^* (\mathbf{x}, t), (\mathbf{x}^*, t^*), D_T)]$	$I(y; \mathbf{y}^* (\mathbf{x}, t), D_T)$
	Latent PO Responses ($f(\mathbf{x}^*, t^*)$)	$\mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)}[\sum_{t^*} I(y; f(\mathbf{x}^*, t^*))]$ (Additive) $\mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)}[I(y; (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1)))]$ (Joint PO)	$I(y; \mathbf{y}_{PO}^* (\mathbf{x}, t), D_T)$
	CATE ($\tau(\mathbf{x}^*)$)	$\mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)}[I(y; \tau(\mathbf{x}^*))]$	$I(y; \tau (\mathbf{x}, t), D_T)$

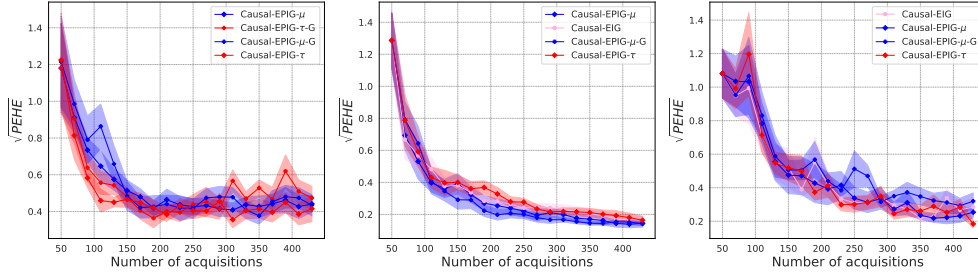


Figure 7. Performance comparison on the Hahn linear dataset under distribution shift. Panels show different base CATE estimators. The scalable mean-marginal variants generally match or improve over their global counterparts in this setting.

in model outputs. Causal-EPIG belongs to the latter family, but replaces factual prediction targets with causal targets. Within prediction-focused methods, factual EPIG targets future factual outcomes, PO-based Causal-EPIG targets the latent conditional potential-outcome responses, and CATE-based Causal-EPIG targets the final estimand $\tau(\mathbf{x}^*)$. The PO target can be formulated additively, $I(y; f(\mathbf{x}^*, 0)) + I(y; f(\mathbf{x}^*, 1))$, or jointly, $I(y; (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1)))$. The joint formulation retains the modeled posterior covariance between the two response surfaces and is the primary Causal-EPIG- μ objective used in the paper; the additive version is denoted Causal-EPIG- μ -S in the appendix experiments. Finally, information can be aggregated over the target population in two ways. The mean-marginal formulation averages pointwise information gains over $\mathbf{X}_{\text{tar}}$ and scales well computationally. The global formulation measures information about the full target vector jointly, capturing inter-target dependencies but incurring substantially higher cost. Our main experiments use the mean-marginal formulation for scalability, while global variants are used for theoretical comparison and ablation.

D.3. Computational Complexity and Runtime Analysis

The main computational cost of Causal-EPIG is determined by the pool size n_P , target-set size $n_{\text{tar}} = |\mathbf{X}_{\text{tar}}|$, and number of posterior samples n_M for sample-based estimators. The mean-marginal formulation used in our experiments scores all candidates in $\mathcal{O}(n_P n_{\text{tar}} n_M)$, scaling linearly in both pool and target-set size. By contrast, the global formulation requires operations on the full target covariance matrix for each candidate, leading to roughly $\mathcal{O}(n_P n_{\text{tar}}^3)$ cost and making it expensive for large target sets. Tab. 4 reports the empirical acquisition time per batch for mean-marginal and global variants. The mean-marginal objectives are substantially faster, with Causal-EPIG- μ being about 20 $\times$ faster than its global version on average. Tab. 5 compares Causal-EPIG- τ with standard baselines. Causal-EPIG is more expensive than simpler BALD-style baselines, but this reflects the intended computational–statistical trade-off: the method spends additional computation to target uncertainty in the causal estimand, which is appropriate when outcome acquisition is the dominant cost.

Table 4. Average acquisition time (seconds) per batch for mean-marginal and global (-G) Causal-EPIG formulations. Results are mean $\pm$ std across trials.

Estimator	Causal-EPIG- μ	Causal-EPIG- μ -G	Causal-EPIG- τ	Causal-EPIG- τ -G
BCF	0.7731 ± 0.0508	16.1392 ± 0.5763	0.5392 ± 0.0342	4.1811 ± 0.1036
CMGP	0.1813 ± 0.0164	4.1814 ± 0.1598	0.4016 ± 0.0289	1.9840 ± 0.0883
NSGP	0.3931 ± 0.0331	7.5014 ± 1.0600	0.3812 ± 0.0334	2.9213 ± 0.4922
Overall	0.4492 ± 0.2999	9.2740 ± 6.1728	0.4406 ± 0.0860	3.0288 ± 1.1025

Table 5. Average acquisition time (seconds) of Causal-EPIG- τ compared with representative baselines.

Methods	Random	μ -BALD	$\mu\rho$ -BALD	$\mu\pi$ -BALD	Causal-EPIG- τ
	(6.5 ± 0.7)	(3.6 ± 0.1)	(9.6 ± 0.4)	(9.6 ± 0.4)	(4.4 ± 0.1)
Time (s)	$\times 10^{-5}$	$\times 10^{-3}$	$\times 10^{-3}$	$\times 10^{-3}$	$\times 10^{-1}$

E. Theoretical Analysis Details

E.1. Framework Mapping to TAL

Our analysis uses the transductive active learning (TAL) framework of [Hübotter et al. \(2024\)](#) to identify the global Causal-EPIG objective, but proves the uncertainty bound directly for our finite, no-replacement pool. Throughout, the potential-outcome targets are the latent conditional response surfaces $f(\mathbf{x}, t) = \mathbb{E}[Y(t) \mid \mathbf{x}]$, rather than additional noisy potential-outcome realizations.

Augmented GP. We model the two potential outcome surfaces $f_t(\mathbf{x})$ as a single GP $f(\tilde{\mathbf{x}})$ over the augmented input space $\tilde{\mathcal{X}} = \mathcal{X} \times \{0, 1\}$, with $\tilde{\mathbf{x}} = (\mathbf{x}, t)$. The augmented kernel $\tilde{k}$ can be written using a Linear Model of Coregionalization (LMC) construction,

$$\tilde{k}((\mathbf{x}, t), (\mathbf{x}', t')) := \sum_{q=1}^Q (B_q)_{t,t'} k_q(\mathbf{x}, \mathbf{x}'), \quad (32)$$

where each k_q is a positive semidefinite kernel and each $B_q \geq 0$ is a 2×2 coregionalization matrix. Hence $\tilde{k}$ is a valid LMC kernel and induces a joint GP over all potential-outcome components. In the CMGP experiments, we use $Q = 2$, following the CMGP specification.

Target and sample spaces. The target space is the augmented set of target potential outcomes, $\mathcal{A} := \tilde{\mathcal{X}}_{\text{tar}} = \{(\mathbf{x}^*, t) : \mathbf{x}^* \in \mathcal{X}_{\text{tar}}, t \in \{0, 1\}\}$. The sample space is the finite set of factual covariate–treatment pairs available for acquisition, $\mathcal{S} := \tilde{\mathcal{D}}_P$. Target variables are latent potential-outcome components in $\mathcal{A}$, whereas admissible queries are noisy factual observations in $\mathcal{S}$. Queried units are removed from the pool; if $B_n = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n\}$, the remaining pool is $\mathcal{S} \setminus B_n$.

Identification with Global ITL. Let D_0 denote a warm-start labeled set and let $D_n = (D_0, y_{B_n})$. Conditioning on D_0 , sequential Causal-EPIG- μ -G selects

$$\tilde{\mathbf{x}}_{n+1} \in \arg \max_{\tilde{\mathbf{x}} \in \mathcal{S} \setminus B_n} \mathbb{I}(f_{\mathcal{A}}; y_{\tilde{\mathbf{x}}} \mid D_n). \quad (33)$$

The candidate utility is exactly the Global ITL objective in [Hübotter et al. \(2024, Eq. 2\)](#). The selection process here is nevertheless pool-based and without replacement. Accordingly, the result below is proved directly for subsets of $\mathcal{S}$ rather than by applying the repeatable-sampling TAL convergence theorem.

E.2. Submodularity: Scope of the Assumption

The finite-pool greedy analysis uses the set function

$$\psi(\tilde{A}) := \mathbb{I}(f_{\mathcal{A}}; y_{\tilde{A}} \mid D_0), \quad \tilde{A} \subseteq \mathcal{S}, \quad (34)$$

with marginal gain

$$\Delta(\tilde{\mathbf{x}} \mid \tilde{A}) := \psi(\tilde{A} \cup \{\tilde{\mathbf{x}}\}) - \psi(\tilde{A}). \quad (35)$$

Assumption 4.2 states that ψ is submodular, i.e., $\Delta(\tilde{x} \mid \tilde{A}') \geq \Delta(\tilde{x} \mid \tilde{A})$ for all $\tilde{A}' \subseteq \tilde{A} \subseteq \mathcal{S}$ and $\tilde{x} \in \mathcal{S} \setminus \tilde{A}$. Mutual information is non-negative, and the chain rule makes ψ monotone.

When the assumption holds. Under the conditionally independent observation-noise model used here, a sufficient condition is $\mathcal{S} \subseteq \mathcal{A}$. In this case, the objective is a standard Gaussian ITL objective over target variables, and submodularity follows from Hübottter et al. (2024, Lemma C.9). In our CATE setting, this condition holds in a no-shift transductive setting where every factual query component in $\tilde{\mathcal{D}}_P$ is included in the augmented target set $\tilde{\mathcal{X}}_{\text{tar}}$.

When the assumption may fail. The assumption is not guaranteed when $\mathcal{S} \not\subseteq \mathcal{A}$, for example under target covariate shift. It may also fail for derived-target objectives, such as utilities involving $\tau(\mathbf{x}) = f(\mathbf{x}, 1) - f(\mathbf{x}, 0)$, or for mean-marginal approximations that do not coincide with the global ITL objective. In such cases, synergistic effects can occur, as in Hübottter et al. (2024, Example C.8). Thus, Assumption 4.2 is a technical condition for the global PO-based finite-pool analysis, not a universal claim for all Causal-EPIG variants.

E.3. Finite-Pool Uncertainty Contraction

Condition on D_0 and regard its posterior GP as the prior for acquisition. Throughout this subsection, the GP kernel, hyperparameters, and independent Gaussian observation-noise model are fixed, with strictly positive noise variances. Deterministic tie-breaking is used. Let $n_P = |\mathcal{S}|$ and define

$$\psi(B) := \mathbb{I}(f_{\mathcal{A}}; y_B \mid D_0), \quad \gamma_k := \max_{B \subseteq \mathcal{S}, |B| \leq k} \psi(B), \quad (36)$$

$$\eta_{\mathcal{D}_P}^2(\tilde{\mathbf{x}}^*) := \text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_0, y_{\mathcal{S}}], \quad \sigma_0^2(\tilde{\mathbf{x}}^*) := \text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_0]. \quad (37)$$

Because ψ is monotone, $\gamma_{n_P} = \psi(\mathcal{S})$.

Lemma E.1 (Full-pool uncertainty identity). *For any fixed queried subset $B \subseteq \mathcal{S}$ and any $\tilde{\mathbf{x}}^* \in \mathcal{A}$ with $\sigma_0^2(\tilde{\mathbf{x}}^*) > 0$,*

$$\frac{1}{2} \log \frac{\text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_0, y_B]}{\eta_{\mathcal{D}_P}^2(\tilde{\mathbf{x}}^*)} = \mathbb{I}(f(\tilde{\mathbf{x}}^*); y_{\mathcal{S} \setminus B} \mid D_0, y_B) \quad (38)$$

$$\leq \psi(\mathcal{S}) - \psi(B). \quad (39)$$

Proof. Strictly positive observation-noise variances imply $\eta_{\mathcal{D}_P}^2(\tilde{\mathbf{x}}^*) > 0$ whenever $\sigma_0^2(\tilde{\mathbf{x}}^*) > 0$. Conditional mutual information between jointly Gaussian variables is the log ratio of their conditional variances, giving the equality. Since $f(\tilde{\mathbf{x}}^*)$ is a component of $f_{\mathcal{A}}$, the chain rule gives

$$\mathbb{I}(f(\tilde{\mathbf{x}}^*); y_{\mathcal{S} \setminus B} \mid D_0, y_B) \leq \mathbb{I}(f_{\mathcal{A}}; y_{\mathcal{S} \setminus B} \mid D_0, y_B) \quad (40)$$

$$= \psi(\mathcal{S}) - \psi(B), \quad (41)$$

where the last equality is another application of the chain rule. $\square$

Proposition E.2 (Explicit finite-pool contraction). *Under the conditions of Thm. 4.3, define*

$$\epsilon_n := \min \left\{ \frac{n_P - n}{n_P} \gamma_{n_P}, \min_{1 \leq k \leq n_P} [(\gamma_{n_P} - \gamma_k) + e^{-n/k} \gamma_k] \right\}. \quad (42)$$

Then, for every $1 \leq n \leq n_P$ and $\tilde{\mathbf{x}}^ \in \mathcal{A}$ with $\sigma_0^2(\tilde{\mathbf{x}}^*) > 0$,*

$$\frac{1}{2} \log \frac{\text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_n]}{\eta_{\mathcal{D}_P}^2(\tilde{\mathbf{x}}^*)} \leq \epsilon_n, \quad (43)$$

$$\text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_n] \leq \eta_{\mathcal{D}_P}^2(\tilde{\mathbf{x}}^*) + 2\sigma_0^2(\tilde{\mathbf{x}}^*)\epsilon_n. \quad (44)$$

If $\sigma_0^2(\tilde{\mathbf{x}}^) = 0$, the variance statement is immediate.*

Proof. Let B_i be the set selected after i sequential acquisitions and write

$$R_i := \psi(\mathcal{S}) - \psi(B_i), \quad \Gamma_i := \max_{s \in \mathcal{S} \setminus B_i} \Delta(s \mid B_i). \quad (45)$$

Submodularity gives the covering bound

$$R_i \leq \sum_{s \in \mathcal{S} \setminus B_i} \Delta(s \mid B_i) \leq (n_P - i) \Gamma_i. \quad (46)$$

Sequential Causal-EPIG- μ -G attains Γ_i , so

$$R_{i+1} = R_i - \Gamma_i \leq \left(1 - \frac{1}{n_P - i}\right) R_i. \quad (47)$$

Since $R_0 = \gamma_{n_P}$, multiplication from $i = 0$ to $n - 1$ yields

$$R_n \leq \gamma_{n_P} \prod_{i=0}^{n-1} \frac{n_P - i - 1}{n_P - i} = \frac{n_P - n}{n_P} \gamma_{n_P}. \quad (48)$$

For the capacity-profile bound, fix $1 \leq k \leq n_P$ and let X_k attain γ_k . Submodularity similarly gives

$$\gamma_k - \psi(B_i) \leq \psi(B_i \cup X_k) - \psi(B_i) \leq \sum_{s \in X_k \setminus B_i} \Delta(s \mid B_i) \leq k \Gamma_i. \quad (49)$$

Therefore,

$$\gamma_k - \psi(B_n) \leq \left(1 - \frac{1}{k}\right)^n \gamma_k \leq e^{-n/k} \gamma_k, \quad (50)$$

and hence

$$R_n \leq (\gamma_{n_P} - \gamma_k) + e^{-n/k} \gamma_k. \quad (51)$$

Taking the minimum over this family and the full-pool bound gives $R_n \leq \epsilon_n$. Lemma E.1 proves the log-variance statement and also the exponential form

$$\text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_n] \leq \eta_{D_P}^2(\tilde{\mathbf{x}}^*) e^{2\epsilon_n}. \quad (52)$$

For the additive form, let $a = \text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_n]$ and $b = \eta_{D_P}^2(\tilde{\mathbf{x}}^*)$. Since $0 < b \leq a \leq \sigma_0^2(\tilde{\mathbf{x}}^*)$ and $a - b \leq a \log(a/b)$,

$$a - b \leq 2\sigma_0^2(\tilde{\mathbf{x}}^*) \epsilon_n. \quad (53)$$

When $\sigma_0^2(\tilde{\mathbf{x}}^*) = 0$, all corresponding variances are zero and the claim is immediate. $\square$

Proof of Thm. 4.3. Let

$$\sigma_{\max}^2 := \max_{\tilde{\mathbf{x}}^* \in \mathcal{A}} \sigma_0^2(\tilde{\mathbf{x}}^*). \quad (54)$$

If $\gamma_{n_P} = 0$, Prop. E.2 has $\epsilon_n = 0$ and the claimed inequality holds for any $C > 0$. Suppose $\gamma_{n_P} > 0$. Normalization and submodularity give

$$\gamma_{n_P} = \psi(\mathcal{S}) \leq \sum_{s \in \mathcal{S}} \psi(\{s\}) \leq n_P \gamma_1, \quad (55)$$

so $\gamma_n \geq \gamma_1 > 0$ for every $1 \leq n \leq n_P$. Moreover, $\gamma_{n_P} > 0$ implies $\sigma_{\max}^2 > 0$. Hence the finite, strictly positive, budget-independent constant

$$C := 2\sigma_{\max}^2 \max \left\{ 1, \max_{1 \leq n \leq n_P} \frac{\sqrt{n} \epsilon_n}{\gamma_n} \right\} < \infty \quad (56)$$

is well defined. The additive conclusion of Prop. E.2 then gives

$$\text{Var}[f(\tilde{\mathbf{x}}^*) \mid D_n] - \eta_{D_P}^2(\tilde{\mathbf{x}}^*) \leq C \frac{\gamma_n}{\sqrt{n}} \quad (57)$$

uniformly over $\tilde{\mathbf{x}}^* \in \mathcal{A}$ and $1 \leq n \leq n_P$. For reference, the coarser explicit choice

$$C = 2\sigma_{\max}^2 \sqrt{n_P} \frac{\gamma_{n_P}}{\gamma_1} \quad (58)$$

is also admissible, since $\epsilon_n \leq \gamma_{n_P}$, $\gamma_n \geq \gamma_1$, and $n \leq n_P$. $\square$

Corollary E.3 (Stale-score batches for global Causal-EPIG- μ -G). *Under the fixed-model, positive-noise, deterministic tie-breaking, and submodularity conditions of Thm. 4.3, let $B_0 = \emptyset$. Suppose that at batch round r , with B_r already queried and $n_r = |B_r|$, the algorithm selects a nonempty batch $Q_r \subseteq \mathcal{S} \setminus B_r$ containing at least one maximizer of the individual global Causal-EPIG- μ -G score $\Delta(s \mid B_r)$, as does stale top- n_b selection, and sets $B_{r+1} = B_r \cup Q_r$. After R completed batches, define*

$$\epsilon_R^{\text{batch}} := \begin{cases} 0, & |B_R| = n_P, \\ \min \left\{ \gamma_{n_P} \prod_{r=0}^{R-1} \left(1 - \frac{1}{n_P - n_r} \right), \min_{1 \leq k \leq n_P} [(\gamma_{n_P} - \gamma_k) + e^{-R/k} \gamma_k] \right\}, & |B_R| < n_P. \end{cases} \quad (59)$$

Then the log and additive conclusions of Prop. E.2 hold with D_n and ϵ_n replaced by (D_0, y_{B_R}) and $\epsilon_R^{\text{batch}}$, respectively.

Proof. Each stale-score batch contains an element attaining $\Gamma_r = \max_{s \in \mathcal{S} \setminus B_r} \Delta(s \mid B_r)$. By monotonicity, the joint batch gain is at least Γ_r . The two covering arguments above therefore give

$$R_{r+1} \leq \left(1 - \frac{1}{n_P - n_r} \right) R_r, \quad (60)$$

$$\gamma_k - \psi(B_{r+1}) \leq \left(1 - \frac{1}{k} \right) (\gamma_k - \psi(B_r)). \quad (61)$$

Iteration gives the displayed product and capacity-profile terms. If $B_R = \mathcal{S}$, then $R_R = 0$ exactly. Lemma E.1 and the same scalar inequality as in Prop. E.2 complete the proof. $\square$

Corollary E.3 applies only when stale batch scores are the marginal gains of the global set utility ψ . The experimental Causal-EPIG configurations use mean-marginal μ - or τ -based scores, for which this identification generally fails; those top- n_b batches are therefore not covered by the corollary.

E.4. Interpretation of Thm. 4.3

The floor $\eta_{D_P}^2$ is the posterior uncertainty attainable after every accessible factual unit has been queried once. It respects the observational and no-replacement constraints of pool-based active CATE estimation. It is generally larger than the noiseless latent-pool quantity

$$\eta_{\text{latent}}^2(\tilde{x}^*) := \text{Var}[f(\tilde{x}^*) \mid D_0, f_{\mathcal{S}}], \quad (62)$$

because $y_{\mathcal{S}}$ is a noisy garbling of $f_{\mathcal{S}}$. At $n = n_P$, the explicit bound in Prop. E.2 reaches $\eta_{D_P}^2$ exactly.

The factor $(n_P - n)\gamma_{n_P}/n_P$ can be loose in the small-budget regime $n \ll n_P$. This limitation is unavoidable without additional structure: for a modular utility in which all pool elements carry equal independent information, the residual information is exactly $(n_P - n)\gamma_{n_P}/n_P$. The capacity-profile term captures a complementary regime. If γ_k is already close to γ_{n_P} for some moderate k , then $(\gamma_{n_P} - \gamma_k) + e^{-n/k} \gamma_k$ can certify substantial contraction long before the pool is exhausted. Thus, the bound is most informative either near pool exhaustion or when the information-capacity curve saturates early.

The main theorem's $C\gamma_n/\sqrt{n}$ display is a compact envelope of this explicit finite-pool result. The constant C is uniform over the finite budget range but may depend on n_P , the fixed pool, the target set, and the capacity profile; the display therefore does not assert an asymptotic $1/\sqrt{n}$ rate under increasing or repeatable access. The substantive budget dependence is the explicit ϵ_n in Prop. E.2.

The theorem controls every component of the augmented target vector, including counterfactual-arm components and targets outside the acquisition pool, subject to Assumption 4.2. It is a model-conditional result: refitting the kernel, hyperparameters, or noise model from newly observed outcomes is outside its scope. It should also not be read as a guarantee for the practical mean-marginal utilities, which are treated separately below.

E.5. Finite-Pool Analysis for Mean-Marginal Causal-EPIG

The global joint-PO objective is computationally expensive, whereas the practical Causal-EPIG strategies use mean-marginal (MM) objectives. Here we identify Causal-EPIG- τ with MM-ITL and give two results that do not invoke an unproved MM

variance analogue: a sequential marginal-gain bound and an exact information–variance identity. We do not claim a uniform average-variance convergence rate.

Induced Gaussian instance. Let $n_{\text{tar}} = |\mathbf{X}_{\text{tar}}|$ and write $\mathbf{X}_{\text{tar}} = \{\mathbf{x}_1^*, \dots, \mathbf{x}_{n_{\text{tar}}}^*\}$. Define

$$z_{\mathbf{x}^*} := \tau(\mathbf{x}^*) = f((\mathbf{x}^*, 1)) - f((\mathbf{x}^*, 0)). \quad (63)$$

Because every $z_{\mathbf{x}^*}$ is a linear functional of the augmented GP, the target variables $\{z_{\mathbf{x}^*}\}_{\mathbf{x}^* \in \mathbf{X}_{\text{tar}}}$ and the pool latent variables $\{f(s)\}_{s \in \mathcal{S}}$ form a valid finite jointly Gaussian target–sample instance.

Identification with MM-ITL. For a candidate $s \in \mathcal{S} \setminus B_n$, the Causal-EPIG- τ utility is

$$\psi_{\text{MM}}^\tau(s \mid D_n) := \frac{1}{|\mathbf{X}_{\text{tar}}|} \sum_{\mathbf{x}^* \in \mathbf{X}_{\text{tar}}} \text{I}(z_{\mathbf{x}^*}; y_s \mid D_n), \quad (64)$$

which induces the same acquisition rule as the MM-ITL objective of [Hübottter et al. \(2024, App. D.3\)](#) on the induced Gaussian instance, up to the positive normalization factor $1/|\mathbf{X}_{\text{tar}}|$.

Let

$$\Gamma_n^{\tau, \text{MM}} := \max_{s \in \mathcal{S} \setminus B_n} \psi_{\text{MM}}^\tau(s \mid D_n) \quad (65)$$

be the maximal remaining MM gain.

Assumption E.4 (Non-increasing MM maximal gains). For the sequential no-replacement rule,

$$\Gamma_{n+1}^{\tau, \text{MM}} \leq \Gamma_n^{\tau, \text{MM}} \quad \text{for } 0 \leq n < n_P - 1. \quad (66)$$

Remark E.5. Assumption E.4 is not automatic in general transductive settings, where synergistic effects may occur ([Hübottter et al., 2024](#)). It is imposed only for the sequential MM gain bound below.

Define the average single-target information capacity

$$\bar{\gamma}_n^\tau := \frac{1}{|\mathbf{X}_{\text{tar}}|} \sum_{\mathbf{x}^* \in \mathbf{X}_{\text{tar}}} \max_{\tilde{B} \subseteq \mathcal{S}, |\tilde{B}| \leq n} \text{I}(z_{\mathbf{x}^*}; y_{\tilde{B}} \mid D_0). \quad (67)$$

Lemma E.6 (Sequential MM marginal-gain decay). *Under the same fixed-model and deterministic tie-breaking conditions as Thm. 4.3, suppose sequential no-replacement Causal-EPIG- τ greedily attains $\Gamma_i^{\tau, \text{MM}}$ at every step and Assumption E.4 holds. Then, for $1 \leq n < n_P$,*

$$\Gamma_n^{\tau, \text{MM}} \leq \frac{\bar{\gamma}_n^\tau}{n}. \quad (68)$$

Proof. The chain rule and sequential greedy selection give

$$\sum_{i=0}^{n-1} \Gamma_i^{\tau, \text{MM}} = \frac{1}{|\mathbf{X}_{\text{tar}}|} \sum_{\mathbf{x}^* \in \mathbf{X}_{\text{tar}}} \text{I}(z_{\mathbf{x}^*}; y_{B_n} \mid D_0) \leq \bar{\gamma}_n^\tau. \quad (69)$$

Assumption E.4 implies that every term in the sum is at least $\Gamma_n^{\tau, \text{MM}}$, proving the claim. $\square$

Lemma E.7 (Information–variance identity for scalar CATE targets). *For any fixed selected subset $B \subseteq \mathcal{S}$ and target $\mathbf{x}^*$ with $\text{Var}[z_{\mathbf{x}^*} \mid D_0] > 0$,*

$$\text{Var}[z_{\mathbf{x}^*} \mid D_0, y_B] = \text{Var}[z_{\mathbf{x}^*} \mid D_0] \exp\{-2\text{I}(z_{\mathbf{x}^*}; y_B \mid D_0)\}. \quad (70)$$

Proof. The variables are jointly Gaussian, so their mutual information is one half of the log ratio between the conditional variances before and after observing y_B . Rearranging gives the identity. The zero-variance case is trivial. $\square$

Lemma E.6 upper-bounds the next maximal *average marginal* information gain; it does not lower-bound the cumulative information acquired about each target. Consequently, it cannot by itself be converted into a uniform average-variance rate via Lemma E.7. Establishing such a rate requires additional control of target-wise information allocation and is left for

future work. The gain lemma is sequential-only; in particular, it does not cover the stale-score top- n_b mean-marginal batches used in the experiments. The variance identity itself holds for any fixed selected subset.

Relation to the μ -based mean-marginal strategy. For each target covariate, define $\mathbf{u}_{\mathbf{x}^*} = (f((\mathbf{x}^*, 0)), f((\mathbf{x}^*, 1)))$. Causal-EPIG- μ is the vector-target analogue of MM-ITL. A corresponding vector-valued convergence rate would require additional analysis. Nevertheless,

$$\text{Var}[\tau(\mathbf{x}^*) \mid D_n] \leq 2\text{Var}[f((\mathbf{x}^*, 1)) \mid D_n] + 2\text{Var}[f((\mathbf{x}^*, 0)) \mid D_n], \quad (71)$$

so marginal PO uncertainty gives a valid, potentially loose, upper bound on CATE uncertainty.

E.6. Implications of the LMC Kernel Structure

The finite-pool bounds depend on the capacity profile $\{\gamma_k\}_{k=1}^{n_P}$, where each γ_k is determined by the joint mutual information $I(f_{\tilde{\mathbf{X}}_{\text{tar}}}; y_{\tilde{B}} \mid D_0)$ and hence by the augmented kernel $\tilde{k}$. Under the LMC construction,

$$\tilde{k}((\mathbf{x}, t), (\mathbf{x}', t')) = \sum_{q=1}^Q (B_q)_{t,t'} k_q(\mathbf{x}, \mathbf{x}'), \quad (72)$$

the off-diagonal entries of B_q encode shared structure between the two potential-outcome surfaces. With diagonal B_q , the capacity reflects learning two separate surfaces. Nonzero off-diagonal entries permit information transfer across treatment arms and may cause the capacity profile to saturate at a smaller k , which tightens the capacity-profile term in Prop. E.2. This is a structural possibility rather than a universal comparison: the actual capacities depend on the kernel, target set, noise levels, and pool geometry.

E.7. Connections to Other Causal Strategies

1. Connection to mean-marginal strategies. The finite-pool theorem applies to the global, joint PO-based Causal-EPIG- μ -G objective. This strategy is theoretically natural because its candidate score is the Global ITL marginal gain over the augmented target vector. The practical Causal-EPIG utilities used in our experiments instead average pointwise information gains over $\mathbf{X}_{\text{tar}}$. App. E.5 provides a sequential marginal-gain bound and an exact scalar information–variance identity for the CATE-targeted MM strategy, but not a full variance-convergence theorem or a stale-batch guarantee.

2. Relationship between PO-based and CATE-based targets. The relationship between the PO-based objective Causal-EPIG- μ and the CATE-based objective Causal-EPIG- τ follows from the data processing inequality. For a fixed target covariate $\mathbf{x}^*$, let $\mathbf{u}_{\mathbf{x}^*} = (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1))$ and $\tau(\mathbf{x}^*) = f(\mathbf{x}^*, 1) - f(\mathbf{x}^*, 0)$. Since $\tau(\mathbf{x}^*)$ is a deterministic function of $\mathbf{u}_{\mathbf{x}^*}$, for any query response $y_{\tilde{\mathbf{x}}}$ and conditioning information D_T ,

$$I(y_{\tilde{\mathbf{x}}}; \mathbf{u}_{\mathbf{x}^*} \mid D_T) \geq I(y_{\tilde{\mathbf{x}}}; \tau(\mathbf{x}^*) \mid D_T). \quad (73)$$

Thus, the PO-based target is more comprehensive in the sense that it retains the full joint potential-outcome information, while the CATE-based target compresses this information to the differenced estimand. This does not imply that one acquisition rule is uniformly better than the other: a query can be highly informative for prognostic or arm-specific outcome variation while being less informative about their contrast. Conversely, when the effect-relevant signal is concentrated in the difference, directly targeting $\tau(\mathbf{x}^*)$ can be more sample-efficient. This is the formal basis for the comprehensiveness–focus trade-off discussed in the main paper.

E.8. Proof of Prop. 4.1

Proof. Here $\epsilon_{\text{PEHE}}^{\mathcal{M}}$ denotes the squared PEHE model risk; the square-root PEHE used in experiments is not the object of this equivalence. Let $\mathcal{F}_s$ denote the information available after s acquisition steps, and let $\mathcal{F}_{s+1} = \mathcal{F}_s \vee \sigma(y)$ denote the information after querying candidate $(\mathbf{x}, t)$ and observing its factual outcome y . We use $\mathbb{E}_{s+1}[\cdot] \equiv \mathbb{E}_{y \sim p(y|\mathbf{x}, t, \mathcal{F}_s)}[\cdot]$ for the pre-posterior expectation over the unknown outcome. For a target covariate $\mathbf{x}^*$, write $Z(\mathbf{x}^*) = \tau(\mathbf{x}^*)$. Under squared loss, the Bayes estimator after observing y is the posterior mean $\hat{\tau}_{s+1}(\mathbf{x}^*) = \mathbb{E}[Z(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]$. The model-based posterior risk at $\mathbf{x}^*$ is $\mathbb{E}[(\hat{\tau}_{s+1}(\mathbf{x}^*) - Z(\mathbf{x}^*))^2 \mid \mathcal{F}_{s+1}] = \text{Var}[Z(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]$. Therefore, the expected model-based PEHE after acquiring $(\mathbf{x}, t)$ is

$$\mathbb{E}_{s+1}[\epsilon_{\text{PEHE}}^{\mathcal{M}}(\hat{\tau}_{s+1})] = \mathbb{E}_{s+1}[\mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)}[\text{Var}[\tau(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]]] \quad (74)$$

$$= \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{E}_{s+1} [\mathbb{V}\text{ar}[\tau(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]]], \quad (75)$$

where the second equality follows by Fubini's theorem. Hence, the one-step model-based acquisition objective is

$$\arg \min_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{E}_{s+1} [\mathbb{V}\text{ar}[\tau(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]]]. \quad (76)$$

For GP models with fixed hyperparameters and Gaussian likelihood, the posterior covariance after adding a candidate input $(\mathbf{x}, t)$ is deterministic given $\mathcal{F}_s$ and the candidate input; it does not depend on the realized value of y . Thus the inner expectation over y can be dropped:

$$\arg \min_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{V}\text{ar}[\tau(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]]. \quad (77)$$

Finally, since $\tau(\mathbf{x}^*) = f(\mathbf{x}^*, 1) - f(\mathbf{x}^*, 0)$, the posterior CATE variance is

$$\mathbb{V}\text{ar}[\tau(\mathbf{x}^*) \mid \mathcal{F}_{s+1}] = \mathbb{V}\text{ar}[f(\mathbf{x}^*, 1) \mid \mathcal{F}_{s+1}] + \mathbb{V}\text{ar}[f(\mathbf{x}^*, 0) \mid \mathcal{F}_{s+1}] \quad (78)$$

$$- 2\text{Cov}[f(\mathbf{x}^*, 1), f(\mathbf{x}^*, 0) \mid \mathcal{F}_{s+1}]. \quad (79)$$

Therefore, the optimal one-step model-based acquisition criterion is to minimize the integrated posterior CATE variance, which depends on both marginal potential-outcome uncertainties and their posterior covariance. $\square$

E.8.1. CONNECTION TO THE CAUSAL-EPIG FRAMEWORK

Prop. 4.1 shows that, under the GP and model-based squared-error assumptions, the natural one-step objective is to minimize integrated posterior CATE variance. Causal-EPIG provides information-theoretic acquisition rules that target the causal quantities determining this variance.

Causal-EPIG- μ (Comprehensive Strategy). This strategy maximizes

$$\arg \max_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{I}(y; (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1)) \mid (\mathbf{x}, t), D_T)]. \quad (80)$$

For Gaussian posteriors, this is equivalent to maximizing the expected reduction in the joint entropy of the potential-outcome pair. Since the CATE variance depends on the marginal variances of $f(\mathbf{x}^*, 0)$ and $f(\mathbf{x}^*, 1)$ as well as their covariance, this objective targets the joint posterior structure from which $\mathbb{V}\text{ar}[\tau(\mathbf{x}^*)]$ is constructed.

Causal-EPIG- τ (Focused Strategy). This strategy maximizes information gain about the estimand itself:

$$\arg \max_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{I}(y; \tau(\mathbf{x}^*) \mid (\mathbf{x}, t), D_T)]. \quad (81)$$

Equivalently, it minimizes the expected posterior entropy of $\tau(\mathbf{x}^*)$. For Gaussian posteriors, $H(\tau(\mathbf{x}^*) \mid D) = \frac{1}{2} \log(2\pi e \mathbb{V}\text{ar}[\tau(\mathbf{x}^*) \mid D])$. Under the GP assumption, the posterior variance after querying $(\mathbf{x}, t)$ is deterministic with respect to the future outcome y , so the Causal-EPIG- τ objective is equivalent to

$$\arg \min_{(\mathbf{x}, t) \in D_P} \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\log \mathbb{V}\text{ar}[\tau(\mathbf{x}^*) \mid \mathcal{F}_{s+1}]]. \quad (82)$$

This is not identical to minimizing the integrated variance in Prop. 4.1; it is the corresponding log-variance, entropy-based uncertainty criterion. Thus, Causal-EPIG- τ is a principled information-theoretic proxy that directly targets posterior uncertainty in the final CATE estimand. Together, Causal-EPIG- μ and Causal-EPIG- τ provide two aligned but distinct acquisition targets: the former preserves the full joint potential-outcome structure, while the latter focuses directly on the differenced causal estimand.

E.9. Relationship between Causal-EPIG and Parameter-Based Acquisition

Proposition E.8. Assume that, for any target covariate $\mathbf{x}^*$, the target causal prediction quantity is a measurable function of the corresponding model parameters. Then the parameter-oriented information utilities upper-bound their prediction-oriented counterparts:

$$(a) \quad \mu\text{-BALD}(\mathbf{x}, t) \geq \text{Causal-EPIG} - \mu(\mathbf{x}, t), \quad (83)$$

$$(b) \quad \text{Causal-EIG}(\mathbf{x}, t) \geq \text{Causal-EPIG} - \tau(\mathbf{x}, t). \quad (84)$$

For models without a distinct CATE-parameter block, part (b) should be read with θ_τ replaced by any parameter or latent representation that determines the CATE process.

Proof. Let $D'_T = D_T \cup \{(\mathbf{x}, t)\}$ denote the acquisition context containing the candidate covariates and treatment, before observing the random outcome y . The relevant utilities can be written as

$$\begin{aligned} \mu\text{-BALD}(\mathbf{x}, t) &:= \mathbb{I}(y; \theta \mid D'_T), \\ \text{Causal-EPIG} - \mu(\mathbf{x}, t) &:= \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{I}(y; (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1)) \mid \mathbf{x}^*, D'_T)], \\ \text{Causal-EIG}(\mathbf{x}, t) &:= \mathbb{I}(y; \theta_\tau \mid D'_T), \\ \text{Causal-EPIG} - \tau(\mathbf{x}, t) &:= \mathbb{E}_{p_{\text{tar}}(\mathbf{x}^*)} [\mathbb{I}(y; \tau(\mathbf{x}^*) \mid \mathbf{x}^*, D'_T)]. \end{aligned}$$

For part (a), conditional on $(\mathbf{x}^*, D'_T)$, the target latent potential-outcome pair is downstream of the model parameters θ . Hence $y \longrightarrow \theta \longrightarrow (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1))$ is a conditional Markov chain. By the conditional data processing inequality,

$$\mathbb{I}(y; \theta \mid \mathbf{x}^*, D'_T) \geq \mathbb{I}(y; (f(\mathbf{x}^*, 0), f(\mathbf{x}^*, 1)) \mid \mathbf{x}^*, D'_T). \quad (85)$$

Since $\mathbf{x}^*$ is sampled independently of (y, θ) conditional on D'_T ,

$$\mathbb{I}(y; \theta \mid \mathbf{x}^*, D'_T) = \mathbb{I}(y; \theta \mid D'_T) = \mu\text{-BALD}(\mathbf{x}, t). \quad (86)$$

Taking expectation over $\mathbf{x}^* \sim p_{\text{tar}}$ gives $\mu\text{-BALD}(\mathbf{x}, t) \geq \text{Causal-EPIG} - \mu(\mathbf{x}, t)$.

For part (b), conditional on $(\mathbf{x}^*, D'_T)$, the CATE prediction $\tau(\mathbf{x}^*)$ is downstream of the CATE parameters θ_τ , or the corresponding CATE-determining latent representation. This gives the conditional Markov chain $y \longrightarrow \theta_\tau \longrightarrow \tau(\mathbf{x}^*)$. Thus,

$$\mathbb{I}(y; \theta_\tau \mid \mathbf{x}^*, D'_T) \geq \mathbb{I}(y; \tau(\mathbf{x}^*) \mid \mathbf{x}^*, D'_T). \quad (87)$$

Again, because $\mathbf{x}^*$ is independent of (y, θ_τ) conditional on D'_T ,

$$\mathbb{I}(y; \theta_\tau \mid \mathbf{x}^*, D'_T) = \mathbb{I}(y; \theta_\tau \mid D'_T) = \text{Causal-EIG}(\mathbf{x}, t). \quad (88)$$

Taking expectation over $\mathbf{x}^* \sim p_{\text{tar}}$ gives $\text{Causal-EIG}(\mathbf{x}, t) \geq \text{Causal-EPIG} - \tau(\mathbf{x}, t)$. $\square$

Equality in the inequalities above holds when the prediction quantity retains all information about the corresponding parameters that is relevant to the candidate outcome. For example, part (b) is tight when $y \perp\!\!\!\perp \theta_\tau \mid \tau(\mathbf{x}^*), \mathbf{x}^*, D'_T$. Otherwise, the parameter-focused objective can contain additional information about parameter directions that do not affect the target causal prediction. This is why the inequalities should not be interpreted as a sample-efficiency guarantee for parameter-based acquisition: they show that parameter-based targets are more comprehensive in an information-theoretic sense, whereas Causal-EPIG deliberately focuses the acquisition target on the causal predictions of interest.

F. Further Experimental Results

This section provides supplementary experimental results. We first report full test-set convergence curves for all benchmark datasets and base CATE estimators. We then provide robustness ablations on warm-start size, unlabeled pool size, acquisition batch size, and the Deep Variational GP estimator used in the original Causal-BALD study.

F.1. Benchmark Convergence Curves

CausalBALD synthetic dataset. Fig. 8 and Fig. 9 show test-set $\sqrt{\text{PEHE}}$ on the CausalBALD synthetic benchmark under regular and target-shift settings.

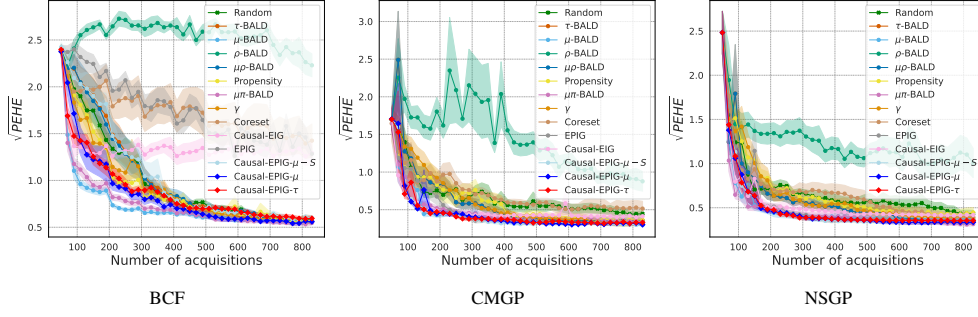


Figure 8. Test-set performance on the CausalBALD synthetic dataset under the regular setup. Each panel reports $\sqrt{\text{PEHE}}$ over the acquisition trajectory for a different base CATE estimator.

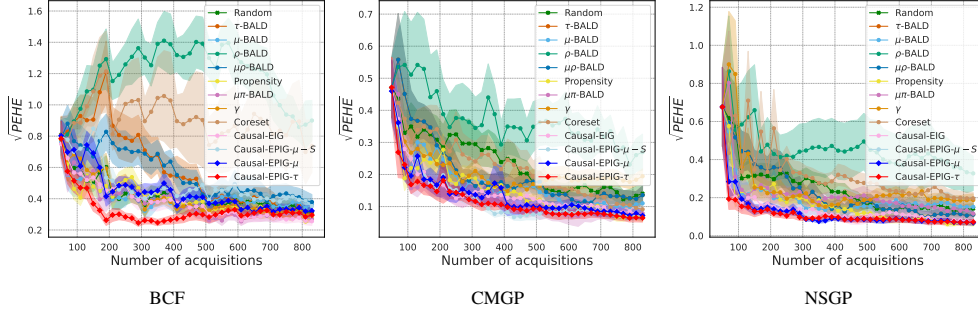


Figure 9. Test-set performance on the CausalBALD synthetic dataset under target distribution shift. Each panel reports $\sqrt{\text{PEHE}}$ over the acquisition trajectory for a different base CATE estimator.

Hahn synthetic dataset. Fig. 10–13 report the corresponding test-set convergence curves for the Hahn linear and nonlinear settings, with and without target distribution shift.

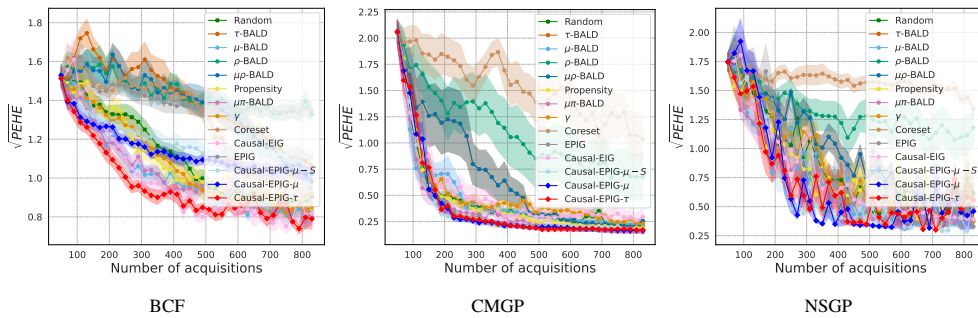


Figure 10. Test-set performance on the Hahn linear synthetic dataset under the regular setup.

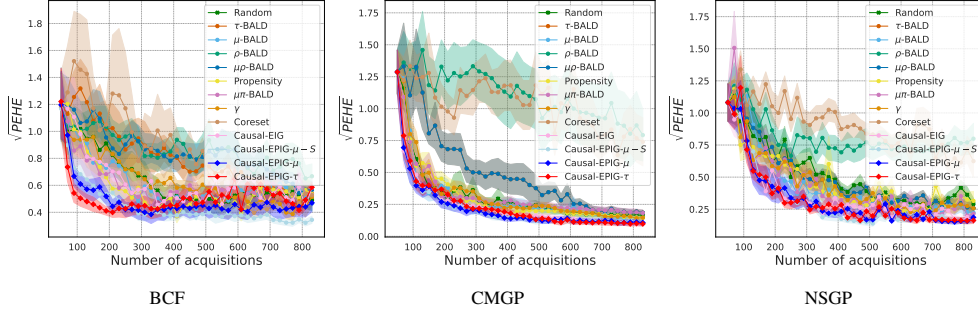


Figure 11. Test-set performance on the Hahn linear synthetic dataset under target distribution shift.

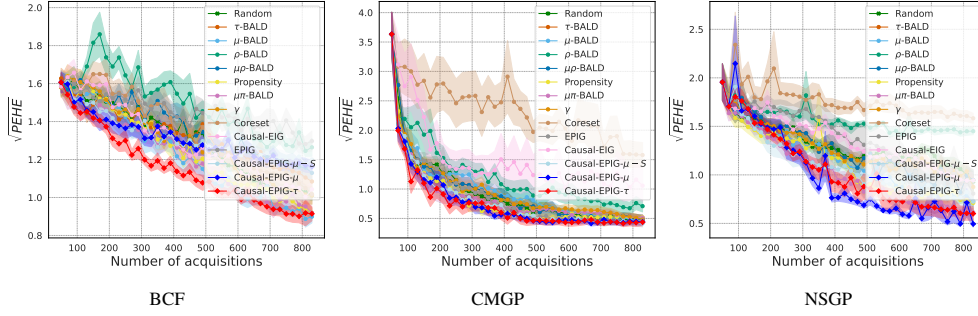


Figure 12. Test-set performance on the Hahn nonlinear synthetic dataset under the regular setup.

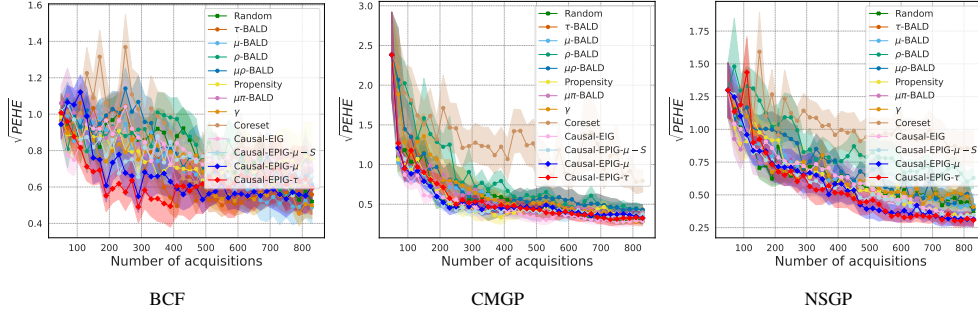


Figure 13. Test-set performance on the Hahn nonlinear synthetic dataset under target distribution shift.

Semi-synthetic datasets. Fig. 14–16 report test-set convergence curves for IHDP and ACTG-175.

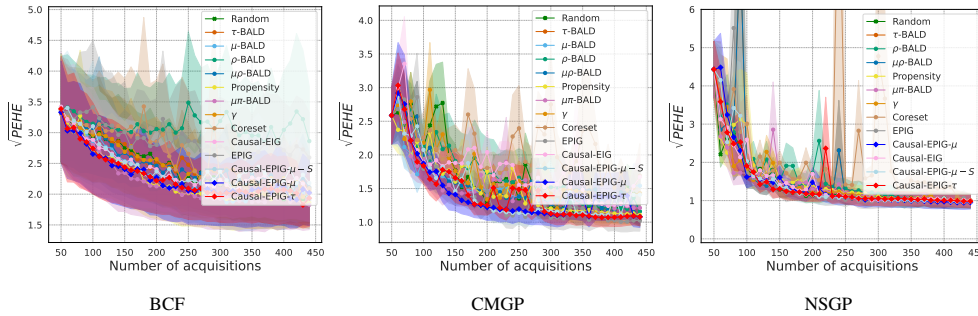


Figure 14. Test-set performance on the IHDP semi-synthetic dataset under the regular setup.

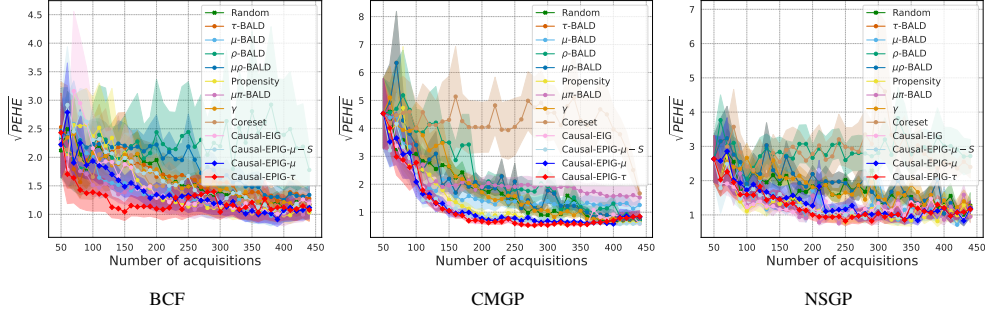


Figure 15. Test-set performance on the IHDP semi-synthetic dataset under target distribution shift.

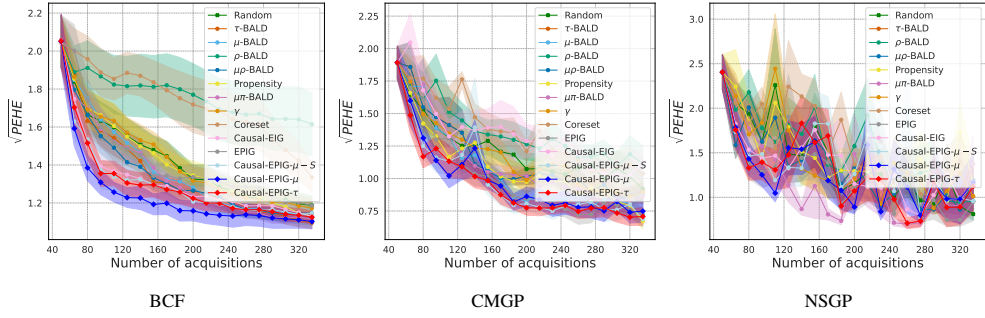


Figure 16. Test-set performance on the ACTG-175 semi-synthetic dataset under the standard setup.

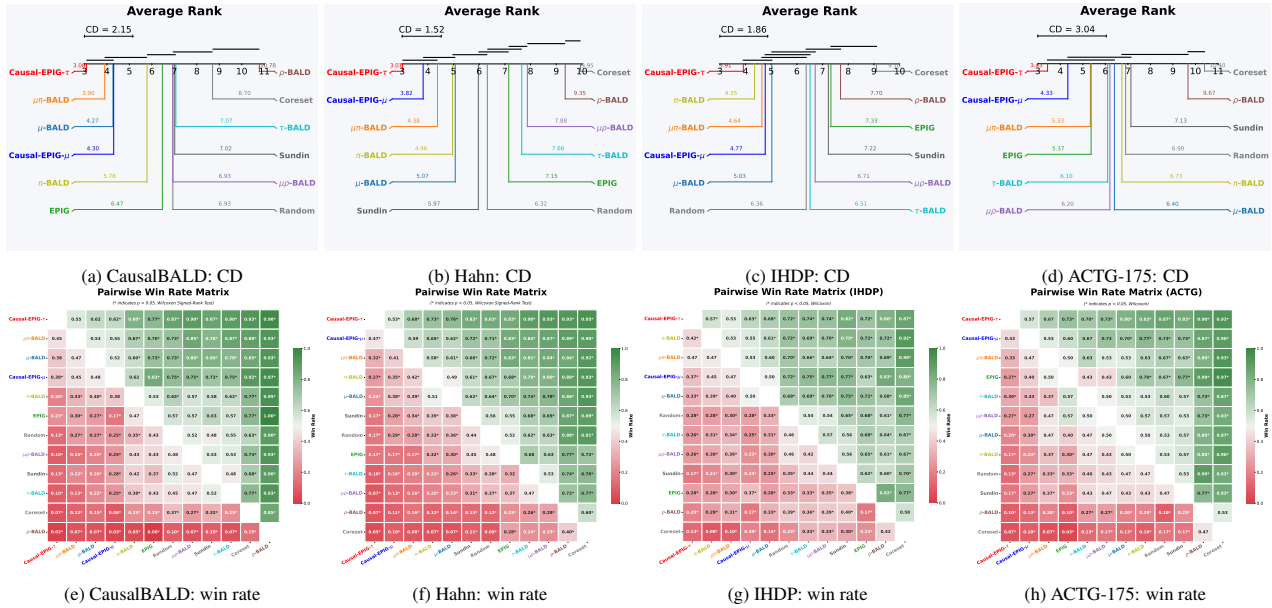


Figure 17. Rank-based comparison of acquisition functions across benchmark groups. The top row shows critical difference (CD) diagrams, and the bottom row shows pairwise win-rate matrices with statistical significance. Columns correspond to the CausalBALD synthetic benchmark, Hahn synthetic benchmarks, IHDP, and ACTG-175.

F.2. Robustness Ablations

We conduct ablations on the Hahn linear dataset with the BCF estimator to assess sensitivity to active-learning hyperparameters. Across these studies, Causal-EPIG maintains strong relative performance, suggesting that its gains are not driven by a single warm-start size, pool size, or batch size.

Warm-start size. Fig. 18 varies the number of initial random samples from 50 to 300. Larger warm starts reduce the initial PEHE for all methods, but the relative ranking remains stable.

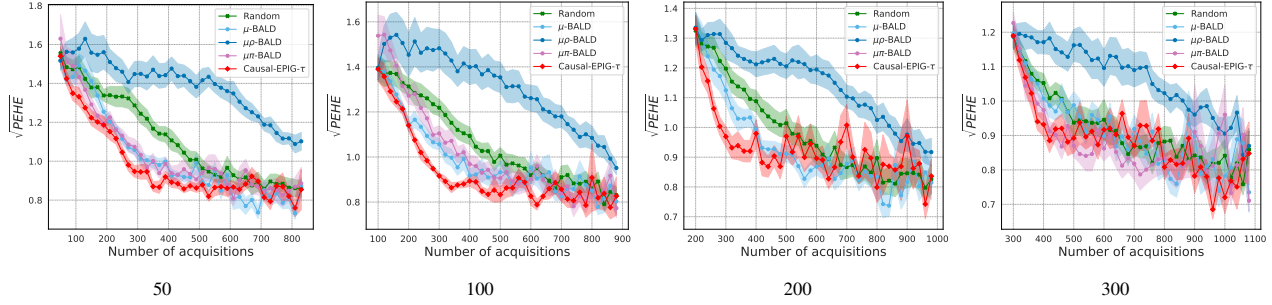


Figure 18. Sensitivity to warm-start size on the Hahn linear dataset with BCF. Panels correspond to 50, 100, 200, and 300 initial random samples.

Pool size. Fig. 19 varies the initial unlabeled pool size from 1000 to 2500. Larger pools give active methods a broader candidate set, but the relative performance trend remains similar.

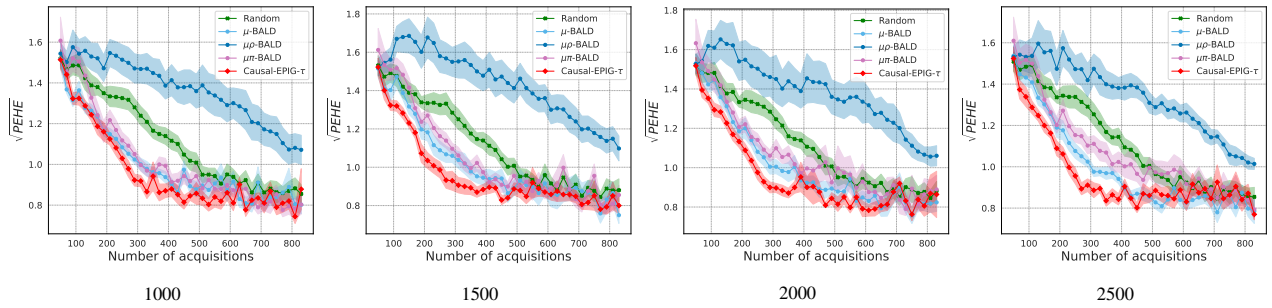


Figure 19. Sensitivity to unlabeled pool size on the Hahn linear dataset with BCF. Panels correspond to $|D_P| = 1000, 1500, 2000, 2500$.

Batch size. Fig. 20 varies the acquisition batch size from 5 to 40. Smaller batches allow more frequent model updates, while larger batches reduce retraining cost; Causal-EPIG remains competitive across this range.

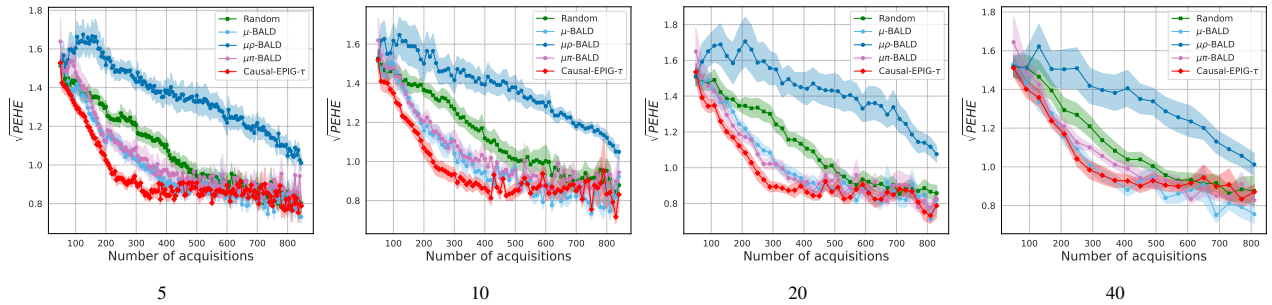


Figure 20. Sensitivity to acquisition batch size on the Hahn linear dataset with BCF. Panels correspond to $n_b = 5, 10, 20, 40$.

Deep Variational GP estimator. To compare with the estimator used in the original Causal-BALD study (Jesson et al., 2021), we also evaluate the Deep Variational GP estimator. Fig. 21 shows that Causal-EPIG remains competitive across these settings, indicating that its behavior is not tied only to BCF or exact GP estimators.

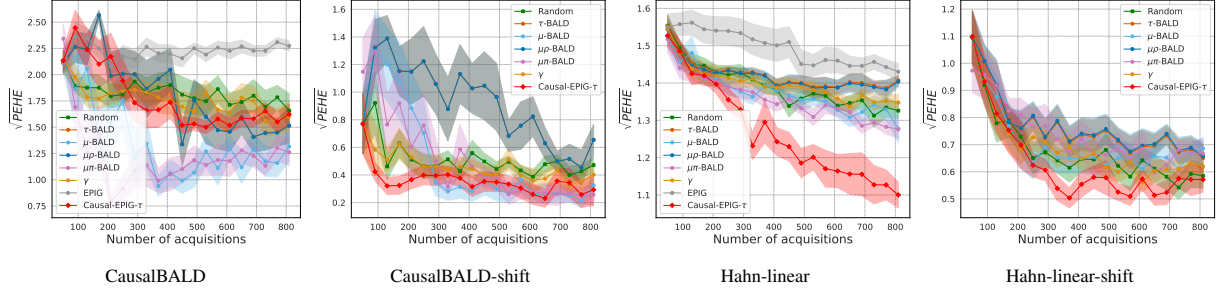


Figure 21. Performance with the Deep Variational GP estimator used in the original Causal-BALD study. Panels show CausalBALD, CausalBALD with shift, Hahn linear, and Hahn linear with shift.

F.3. Targeted Synthetic Study for Causal-EPIG- μ vs. Causal-EPIG- τ

We include a targeted synthetic study to isolate when the PO-based and CATE-based Causal-EPIG objectives are preferred. The covariates are partitioned into prognostic, predictive, and noise blocks. We vary the predictive share of the active informative block, $\rho = \frac{k_{\text{pred}}}{k_{\text{prog}} + k_{\text{pred}}}$, while keeping the total active informative dimension fixed. Thus, changing ρ changes the organization of the informative structure rather than simply changing the number of active variables. We compare two treatment-effect formation mechanisms. In the *surface-split* regime, the effect is induced by the difference of two arm-specific outcome surfaces. Schematically, $\mu_0(x) = g(p) + c(z_C)$ and $\mu_1(x) = g(p) + t(z_T)$, so that $\tau(x) = t(z_T) - c(z_C)$. Here, the treatment-effect signal is distributed across the two potential-outcome surfaces, so learning the joint PO structure remains useful. In the *cancellation* regime, a substantial part of the full-outcome structure is informative for potential-outcome prediction but cancels, or becomes weakly relevant, in the treatment-effect difference. A schematic form is $\mu_0(x) = g(p) + s(x) - \frac{1}{2}r(x)$ and $\mu_1(x) = g(p) + s(x) + \frac{1}{2}r(x)$, so that the differenced target is dominated by the residual effect-bearing signal $r(x)$. In this regime, directly targeting the CATE can be more efficient. The acquisition trajectories are shown in Fig. 5. The surface-split setting favors Causal-EPIG- μ , while the cancellation settings increasingly favor Causal-EPIG- τ as ρ grows. These results support the interpretation that the relevant factor is not merely the number of predictive or prognostic variables, but how the treatment-effect signal is organized relative to the full outcome structure.